



**DESIGN AND IMPLEMENTATION OF A NAVIGATION
FRAMEWORK FOR STEERING CONTROLLED MOBILE
ROBOTS**

MR. NATTAPONG PROMKAEW

**A THESIS FOR THE DEGREE OF MASTER OF ENGINEERING
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**A THESIS SUBMITTED IN PARTIAL FULFILLMENT OF THE
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ณัฐพงศ์ พรหมแก้ว. 2568. การออกแบบและพัฒนาระบบโครงสร้างการนำทางสำหรับหุ่นยนต์เคลื่อนที่ที่ควบคุมด้วยระบบพวงมาลัย. วิทยานิพนธ์ปริญญาวิศวกรรมศาสตรมหาบัณฑิต สาขาวิชาวิศวกรรมเกษตร บัณฑิตวิทยาลัย มหาวิทยาลัยขอนแก่น.

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บทคัดย่อ

การนำทางอย่างแม่นยำในสภาพแวดล้อมกลางแจ้งจำเป็นต้องอาศัยการผสมผสานข้อมูลจากเซนเซอร์หลายชนิดเพื่อเพิ่มประสิทธิภาพของการระบุตำแหน่งและการติดตามเส้นทางการเคลื่อนที่ งานวิจัยนี้นำเสนอการปรับปรุงอัลกอริธึม Pure Pursuit (PP) โดยพัฒนาเป็นเวอร์ชันใหม่เรียกว่า Pure Pursuit with Dynamic Steering Control (PP-DSC) ซึ่งสามารถปรับระยะมองล่วงหน้า (lookahead distance) และความเร็วของหุ่นยนต์ได้แบบปรับตามค่ามุมเลี้ยวแบบไดนามิก อัลกอริธึมนี้ถูกติดตั้งบนหุ่นยนต์เคลื่อนที่อัตโนมัติแบบสตีลล์เลี้ยว (steering-type AMR) โดยใช้แพลตฟอร์ม Robot Operating System 2 (ROS 2) รุ่น Jazzy ข้อมูลเชิงเวลาจริงถูกหลอมรวมจากระบบกำหนดตำแหน่งผ่านดาวเทียมแบบ GNSS-RTK ร่วมกับชุดตรวจวัดความเร่งและการหมุน (IMU) และตัวเข้ารหัสล้อ (wheel encoders) เพื่อเพิ่มความแม่นยำในการระบุตำแหน่ง การทดลองดำเนินการบนเส้นทางรูปแบบตรง วงกลม และเลขแปด (figure-eight) ที่ความเร็วระหว่าง 1.0 ถึง 5.0 เมตรต่อวินาที ผลการทดลองแสดงให้เห็นว่า PP-DSC สามารถลดค่าเบี่ยงเบนด้านข้างเฉลี่ยได้เหลือเพียง 0.05 เมตรในเส้นทางตรง และต่ำกว่า 0.08 เมตรในเส้นทางโค้ง ในขณะที่อัลกอริธึม PP มาตรฐานมีค่าเบี่ยงเบนสูงสุดถึง 0.44 เมตร ผลการวิจัยยืนยันว่า PP-DSC ช่วยเพิ่มความแม่นยำในการติดตามเส้นทางและเสถียรภาพของการเคลื่อนที่ โดยเฉพาะอย่างยิ่งในช่วงที่มีการเปลี่ยนทิศทางการเคลื่อนที่อย่างต่อเนื่อง จึงเหมาะสมอย่างยิ่งสำหรับการนำทางอัตโนมัติแบบเรียลไทม์ภายใต้สภาพเส้นทางและพลศาสตร์การเคลื่อนที่ที่หลากหลาย

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ABSTRACT

Accurate navigation in outdoor environments requires integrating multiple sensor sources to improve localization and trajectory tracking performance. This study proposes a modified version of the Pure Pursuit (PP) algorithm, called Pure Pursuit with Dynamic Steering Control (PP-DSC), which adaptively adjusts both the lookahead distance and velocity based on the steering angle. The algorithm was deployed on a four-wheeled steering-type autonomous mobile robot (AMR) using the Robot Operating System 2 (ROS 2) Jazzy framework. Real-time data were fused from the Global Navigation Satellite System with Real-Time Kinematic corrections (GNSS-RTK), an Inertial Measurement Unit (IMU), and wheel encoders to enhance localization accuracy. Experiments were conducted on straight, circular, and figure-eight trajectories at speeds ranging from 1.0 to 5.0 meters per second. Results showed that PP-DSC reduced the mean lateral deviation to 0.05 meters on straight paths and below 0.08 meters on curved paths, whereas the standard PP algorithm exhibited maximum deviations up to 0.44 meters. The findings confirm that PP-DSC improves trajectory tracking accuracy and motion stability, particularly during directional transitions, making it well-suited for real-time autonomous navigation under varied terrain and motion dynamics.

**This dissertation is humbly dedicated to my respected
parents, benefactor, friends, and advisor, who have
consistently provided support, affection, and assistance to
me, motivating me to believe that I could successfully
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CHAPTER I

INTRODUCTION

1.1 Introduction and background

A robot is a programmable machine capable of performing various tasks through predetermined movements, and its operations are pre-planned to execute specific functions. In the context of agricultural mobile robots, the ability to autonomously navigate in diverse environments is crucial. Navigation involves directing the robot's movement from one point to another, employing various sensors suitable for different environments such as indoor and outdoor settings. Various navigation techniques, including artificial intelligence, are employed. Robot navigation refers to a mobile robot's capability to determine its own location, map a path to a target location, and efficiently plan routes. Effective path planning relies on a combination of localization and resolving the robot's current state and the target location within the same coordinates. Mapping involves notations describing the robot's position with reference to its surroundings. The movement of the robot from one point to another is facilitated by navigation, where the robot autonomously decides its current position and maps a path to the target location. The efficiency of path planning depends on the sensor systems applied to the robot, considering different environmental conditions. For instance, utilizing Light Detection and Ranging (LiDAR) in open spaces may not be as suitable as Global Positioning System (GPS) and Real-time kinematic positioning (RTK) due to the vast and unobstructed nature of the environment.

The field of robotics in agriculture is a widely researched area nowadays due to its potential in developing automated systems suitable for agricultural tasks and addressing challenges in farming practices. This includes issues like labor shortages, increased labor costs, and the need for precision in agricultural operations. The objective of research in this field is to develop autonomous machinery systems that can work independently without human control, achieving higher precision and efficiency [1–4]. Various operational aspects need attention, covering elements like localization, navigation, sensing, actuation, and communication.

The operation of the robot requires the use of various sensors, including satellite positioning systems, laser sensors, and inertial measurement units (IMU). The utilization of LiDAR sensors is highly popular in the navigation of robotic systems due to their accuracy and diverse measurement range. However, there are still many limitations in the use of LiDAR sensors, such as the distance between objects and the sensor [5,6].

In the contemporary context, there might be instances of employing the Pixhawk control board for certain tasks, but it is perceived to lack flexibility in development. Therefore, it is envisaged to utilize The Robot Operating System (ROS2), which is a versatile and open-source platform crafted to assist in the development of robotic applications [7]. ROS encompasses an extensive array of functionalities, such as hardware abstraction, device drivers, libraries, visualizers, message-passing, and package management. This system parallels other robot frameworks like Player, YARP, Orocos, CARMEN, Orca, MOOS, and Microsoft Robotics Studio in terms of capabilities.

The primary aim of ROS2 is to promote the reuse of code in the field of robotics research and development. This objective has led to its swift adoption by a multitude of research institutions and companies as their preferred framework for development. ROS2 finds its application across a diverse spectrum of robotic domains, including aerial and ground robots, humanoids, and underwater vehicles. Notable implementations involve humanoid robots such as PR-2 and REEM-C, ground robots like the Clear Path platforms, and various aerial platforms developed by Ascending Technologies.

Moreover, ROS2 provides substantial support for a wide range of popular robotic sensors, including Inertial Measurement Units, GPS receivers, cameras, and lasers. These sensors are all readily accessible through extensive driver support. Despite its foundation in Linux and the absence of guarantees for hard real-time system properties, ROS2 is proficient in managing soft real-time applications and seamlessly integrates with other prominent open-source libraries, including OpenCV, Point Cloud Library, and the Gazebo simulator.

The runtime architecture of ROS2 is characterized by a peer-to-peer network of processes, which may be distributed across multiple machines. These processes

communicate using a variety of methods, including asynchronous, synchronous, and data storage communications, all underpinned by ROS2 robust communication infrastructure.

In contemporary agricultural applications, robots have become increasingly employed to perform repetitive and hazardous tasks across a diverse range of indoor and outdoor environments [8,9]. Despite their growing adoption, achieving reliable autonomous navigation remains a substantial challenge, particularly in complex settings such as greenhouses where Global Positioning System (GPS) signals may be unavailable [10]. To successfully traverse from a designated origin to a defined destination, agricultural robots must integrate several essential capabilities, including robust localization, environmental mapping, effective path planning, and obstacle avoidance [11]. Among these, the ability to accurately establish their position and generate a globally consistent real-time map of the surroundings is paramount for enhancing overall performance. Consequently, there is a critical need for an efficient and precise real-time localization and mapping method to meet the stringent demands of autonomous navigation in agricultural environments [12].

In response to these limitations, the research community has increasingly turned to sensor fusion strategies to enhance robust localization. By integrating Global Navigation Satellite System Real-Time Kinematic (GNSS-RTK) data, which offers centimeter-level precision under ideal conditions, with additional sensor modalities, researchers have achieved more reliable and accurate navigation [13]. Inertial Measurement Units (IMUs) provide continuous orientation, angular velocity, and acceleration data, thereby mitigating reliance on intermittent external references. Visual odometry, leveraging camera-derived feature tracking, further augments positioning estimates in environments where GNSS signals may be weak or unstable. Beyond these modalities, Light Detection and Ranging (LiDAR) sensing introduces rich three-dimensional spatial information, ensuring continuous localization performance even during temporary GNSS outages [14,15].

Recent studies have proposed a variety of approaches to improve localization and tracking accuracy in autonomous systems. These include the integration of GNSS-RTK with MEMS-IMU for better tilt compensation[16]., improvements in planting

precision through sensor fusion[17]., and the application of Pure Pursuit with both fixed and dynamic lookahead adjustments[18-19]. Some methods incorporate fuzzy control [20] or advanced model-based controllers[21] to reduce lateral deviation, particularly in curved or variable environments. Across these works, dynamic lookahead techniques that adjust based on vehicle speed or path curvature have shown significant error reduction, making them especially effective for autonomous agricultural vehicles and unmanned ground systems operating in real-world scenarios.

Furthermore, research has demonstrated that combining GNSS-RTK, microelectromechanical systems (MEMS)-based IMUs, and stereo visual odometry significantly enhances localization robustness in complex agricultural environments [22]. Similarly, using IMUs, differential GPS, and visual odometry within an Extended Kalman Filter (EKF) framework has shown reliable performance where standalone GNSS signals are insufficient [23,24]. These techniques are particularly critical in regions like Japan, where aging agricultural workforces, labor shortages, and increased accident risks demand autonomous solutions with robust localization capabilities. The relevance of these sensor integration strategies also extends beyond agriculture. Autonomous ground vehicles, unmanned aerial systems, and service robots operating in industrial or urban environments also benefit from sensor redundancy and data fusion. When paired with advanced path planning algorithms (e.g., A-Star) and trajectory-tracking controllers (e.g., Pure Pursuit), these systems deliver improved positional accuracy and robustness in dynamic environments [25]. These approaches help mitigate localization drift and ensure safer and more reliable autonomy in real-world applications [26].

This paper is organized as follows: Section 2 provides a detailed discussion of the system design, including hardware and software components, data communication protocols, and the implementation of navigation algorithms, with a focus on improving the Pure Pursuit algorithm for higher-speed operation. Section 3 presents the experimental results conducted in three types of paths: line path, loop path, and figure-eight path. The evaluation is based on data collected from GNSS-RTK, IMU, and encoder sensors to assess tracking accuracy. Section 4 concludes by comparing the performance of the proposed system with the traditional Pure Pursuit algorithm.

1.2 Objective of the research

1.2.1 To design and implement an advanced control system that enables precise positioning and maneuvering of robotic units.

1.2.2 To develop a navigation system for dynamic and wide-open environmental conditions.

1.3 Scope and limitation of the study

1.3.1 Scope of the research

1.3.1.1 The system operates effectively in open field environments with smooth surfaces and no sky obstructions.

1.3.1.2 The control system achieves a positioning error within ± 10 centimeters along the X-axis and ± 15 degrees in Yaw (rotation around the Y-axis) by integrating sensor data from the Inertial Measurement Unit (IMU), Global Navigation Satellite System (GNSS), and Real-Time Kinematics (RTK).

1.3.1.3 The aim is to conduct a comparative analysis of the precision in path-tracking efficacy, encompassing the traversal of linear trajectories, navigation along curvilinear paths, and execution of figure-eight formations.

1.3.1.4 The display component can illustrate the position of the robot.

1.3.2 Limitation of the research

1.3.2.1 Areas with GPS signal obstructions or dense cloud cover.

1.3.2.2 The transmission of robot positional data to the display unit should be conducted within the same network infrastructure

1.3.2.3 The robot cannot circumvent obstacles while operating in autonomous mode.

1.4 Expected benefits

1.4.1 Acquired a robot capable of operating independently in vast open field areas.

1.4.2 Acquired a control and display system for the operational control of the robot.

1.4.3 Acquired pertinent knowledge in robot navigation and positional identification, with the potential for further development in the future.

CHAPTER II

LITERATURE REVIEWS

This chapter discusses a review of the literature and related research used in the research, including:

2.1 Literature related to agricultural robot.

2.2 Literature related to autonomous mobile robot.

2.3 Literature related to global navigation satellite system (GNSS)

2.1 Literature related to agricultural robot

2.1.1 Exploiting the Internet Resources for Autonomous Robots in Agriculture

The incorporation of autonomous robots into agricultural practices represents a pivotal development in the evolution of modern farming. In their 2018 paper, *Exploiting the Internet Resources for Autonomous Robots in Agriculture*, examine this progression, emphasizing the role of Internet technologies such as the Internet of Things (IoT) and cloud computing in augmenting the functionality of these robots [27]. The paper highlights various applications, including the development of a cost-effective, safe, and environmentally friendly pesticide spraying system and the deployment of autonomous orchard vehicles in the USA for tasks like mowing, pruning, training, spraying, blossoming, fruit thinning, harvesting, and sensing [28]. In Europe, the RHEA fleet (see Figure 1), comprising three tractors that collaborate in pesticide application, is another notable example of this technological advancement. Additionally, specialized robotic systems like LadyBird in Australia and Vibro Crop Robotti in Europe demonstrate further innovations [29]. LadyBird, equipped with thermal and infrared detection systems, hyperspectral cameras, stereovision cameras, LIDAR, and GPS, is designed for crop evaluation. The Vibro Crop Robotti emphasizes meticulous seed planting and the mechanical purification of row crops. It utilizes computing systems that vary from centralized to basic distributed designs, facilitating the control of a small selection of sensors and the operation of fundamental farming implements.



Figure 1 (a) Robots in agriculture adapted from traditional vehicles;
(b) Agricultural robots transformed from standard vehicles. [30]

The integration of advanced Internet technologies is revolutionizing agricultural robotics. This integration involves using cloud services for data storage (see Figure 1), scalability, and data analytics. These technological advancements enable robots to perform more efficiently and accurately, leading to improved agricultural practices [31].

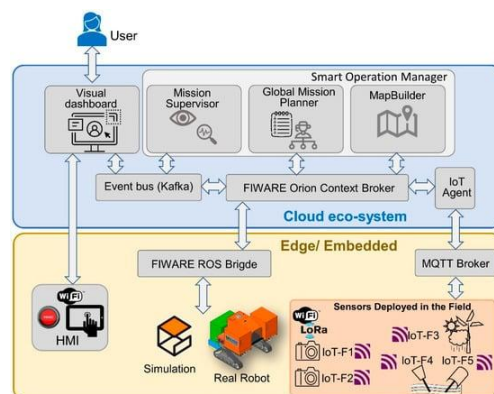


Figure 2 Cloud computing modules/containers [32].

A practical application of this integration is demonstrated through an innovative weeding application using laser technology. This case study exemplifies the

potential of Internet-integrated robotics in agriculture, showing how they can effectively tackle specific farming tasks with enhanced precision.

The architecture proposed in the paper is based on the ROS2 and FIWARE (Future Internet WARE). ROS2 facilitates effective communication among robot subsystems, while FIWARE accelerates the development of smart solutions. This combination is pivotal in realizing the full potential of Internet resources in agricultural robotics.

The study's analysis of transfer speed, latency, and processing time indicates a positive outcome for the use of ROS2 and FIWARE. This integration not only benefits developers by simplifying the development process but also promises to bring significant advancements in agricultural practices.

The integration of Internet resources in agricultural robots signifies a leap forward in precision agriculture. It enhances the efficiency, accuracy, and sustainability of farming practices, paving the way for a more technologically advanced agricultural future.

2.1.2 Low-Cost Robot for Agricultural Image Data Acquisition

The paper "Low-Cost Robot for Agricultural Image Data Acquisition" [32] presents a comprehensive study on the development and application of a low-cost autonomous robot, named DARob, for image data acquisition in agriculture. This essay will review the key aspects of the paper, integrating insights and findings from the literature cited within the document.

The core contribution of the paper is the design and construction of DARob, a low-cost and efficient solution for agricultural image data acquisition. The robot is built with economical components, costing around \$850, making it a feasible option for research groups and applications in developing countries. Its modular design facilitates easy transportation and adaptability to different crop types and sizes. DARob's capability for automatic operation and remote control enhances the repeatability and accuracy of data collection, crucial for precision agriculture.

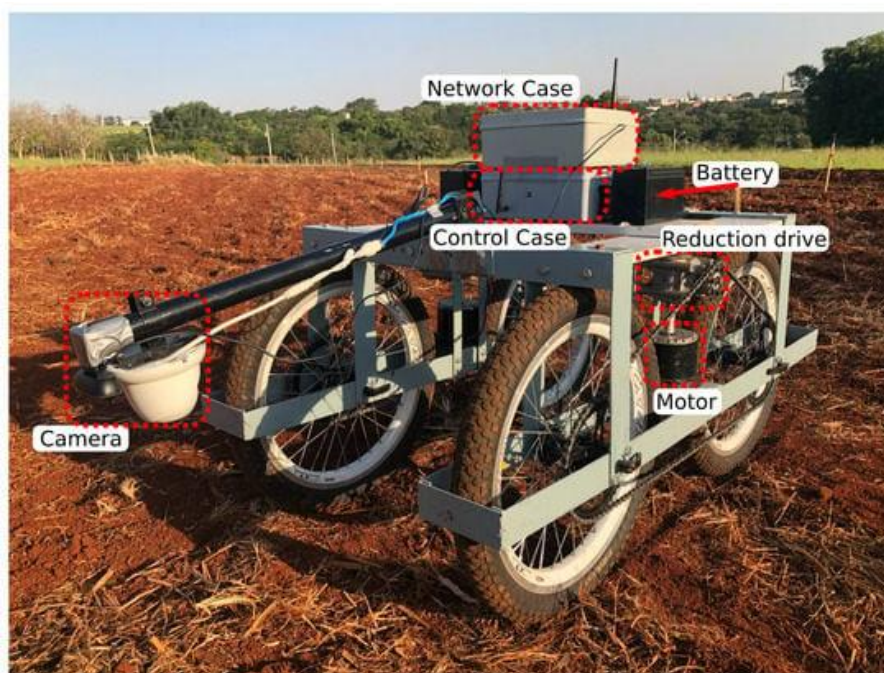


Figure 3 View of the DARob with its main systems [33].

The emergence of agricultural robotics, as investigated by [34-35], and additional researchers, has brought a paradigm shift in agricultural methodologies. Robots like DARob are tailored for specialized agricultural tasks such as weed management, crop surveillance, and precision farming, proving to be more effective than traditional practices. The integration of cutting-edge sensor technologies and artificial intelligence into these robotic systems has remarkably improved their performance, a development underscored in the works of [36-37]. Notwithstanding the substantial advancements in agricultural robotics, the implementation of these technologies encounters notable challenges, primarily due to their high cost and complexity. DARob, which is notable for its cost-effective design, seeks to address these concerns. However, as acknowledged by its creators, DARob confronts specific limitations such as limited battery autonomy and difficulties in navigating muddy terrains. These particular challenges are indicative of broader issues prevalent in the domain of agricultural robotics, as delineated in the research conducted by [38-39], highlighting the imperative for continued research and development in this field.

The integration of DARob into the realm of precision agriculture marks a significant progression. The scholarly work delineates DARob's capability in acquiring high-resolution image data, which proves essential for the segmentation of crops and

weeds, an integral aspect of proficient agricultural administration. This functionality is in concordance with the findings of academics like [40-41]. These scholars underscore the imperative of acquiring accurate data for efficacious crop management and elucidate the enhancement of these processes through the application of machine learning techniques.

2.1.3 Electromagnetic Sensing for Non-Destructive Real-Time Fruit Ripeness Detection: Case-Study for Automated Strawberry Picking

This study focuses on developing a tool for real-time analysis of strawberry ripeness using the Thorvald II robotic system, adaptable to various agricultural needs. The robot (Figure 4), equipped with advanced navigation and localization tools, has shown promise in applying UV light to strawberries for mildew research [42]. Additionally, microwave spectroscopy is explored for its potential in non-destructive, hygienic food analysis [43], aiming to enhance the efficiency of robotic agricultural applications. This technology's versatility has been recognized in various sectors, including healthcare and construction, for its non-invasive evaluation capabilities.



Figure 4 Thorvald II agricultural robotic system [43].

During the harvest season in Drøbak, Norway, in June 2018, Nobel strawberries were selected for picking. The selection process utilized a visual grading system, categorizing strawberries from grades 1 to 4, as detailed in a specific table 1. Since the collection occurred early in the season, no over-ripe fruits were noted, and grade 5 strawberries were obtained by allowing grade 4 fruits to further ripen at room

temperature for about three days post-harvest. This resulted in a minor weight loss for grade 5 strawberries, attributed to drying, approximately 1g per fruit or 6% from their initial weight. The color variation across the grades was documented in figure 5, with the count and total weight of strawberries per grade recorded before measurement.

Table 1 Fruit grading information.

Grade (G)	Grade Description *	n	Total Grade Weight (g)	Mean Fruit Weight (g)
1	Green; entirely not ripe	24	224	9.3
2	Partially ripened with less than 50% of the fruit's surface area showing red coloration.	15	226	15.1
3	More than 50% of the fruit's surface is red, indicating its nearing ripeness, yet it still displays considerable green areas.	11	249	13.5
4	Fully ripe; the entire surface of the fruit has turned red.	14	241	17.2
5	Excessively ripe; characterized by a deep red hue and a noticeable increase in softness. Group 4 fruits underwent a 3-day period at 20°C to reach this over-ripe condition.	14	227	16.2



Figure 5 G1–G4 fruits, indicating variation in color of fruits picked [43].

The examination of spectral data within the 200 – 600 MHz frequency range revealed distinct signal differentiation. Consequently, the analysis concentrated on this frequency spectrum. A Z-score filtering process excluded approximately 9% of the collected data, with aggregated signal amplitudes and their standard deviations detailed in a specific table 2. The analysis enabled the formulation of a linear correlation, achieving an adjusted R² value of 0.788. Regression analysis yielded a p-value of 0.0283, underscoring the statistical significance of these findings, given the threshold of $p < 0.05$.

Table 2 Descriptive statistics from spectra between 200 and 600 MHz [43].

Grade	Total Measure- Ments	Total z-Score Rejections	Mean signal Amplitude (dBm)	Standard Deviation
1	63	3	-0.664	0.422
2	45	5	-1.11	0.576
3	33	3	-1.16	0.917
4	42	3	-1.17	0.866
5	42	9	-1.54	0.640
Air **	21	0	0.000443	0.0125
Total	246	23	-	-

** Air measurements serve as control data to verify the proper functioning of the sensor and system. They also help identify any unexpected variations in system performance, which could be attributed to factors like drift or sensor contamination.

2.2 Literature related to autonomous mobile robot.

2.2.1 Autonomous mobile robot

The introductory section of the paper addresses the intricacies faced by autonomous mobile robots in navigating unpredictable environments, highlighting the critical necessity for continuous navigation and obstacle avoidance. These robots employ a spectrum of sensors, including floor plans, sonar sensing, and Inertial Measurement Units (IMUs) [44], to maneuver both indoor and outdoor settings. Achieving tasks without human intervention demands a diverse range of environmental

sensors, integrated within the robot or externally positioned, contributing to the system's intricate design owing to the requisite utilization of multiple sensors. The paper critically evaluates these challenges and presents an in-depth analysis of devices, sensors, and fusion methodologies aimed at resolving concerns regarding localization, estimation, and navigation, while offering recommendations for further research to enhance the efficacy of autonomous mobile robots.

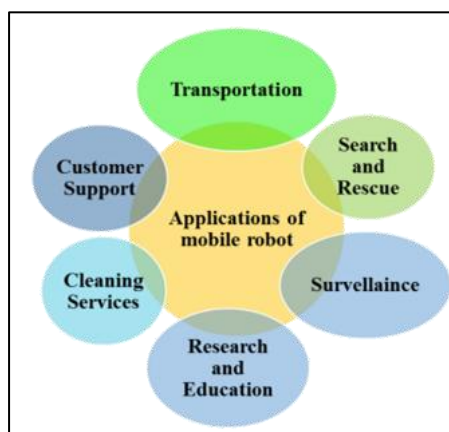


Figure 6 Applications of mobile robot [44].

2.3 Literature related to global Navigation Satellite System (GNSS)

2.3.1 Robot positioning and navigation technology is based on Integration of the Global Navigation Satellite System and real-time kinematics

The essay comprehensively examines the enhancement of robotic positioning and navigation by amalgamating Global Navigation Satellite System (GNSS) and Real-Time Kinematics (RTK) technologies (see figure 7), with a specific focus on surmounting the challenges faced in outdoor environments, where conventional visual Simultaneous Localization and Mapping (SLAM) systems are often inadequate. It underscores the criticality of precise and dependable navigational capabilities for mobile robots across diverse applications. The essay delves into the limitations inherent to current methodologies and articulates the advantages conferred by the synergy of GNSS-RTK and visual SLAM. This integrative approach is shown to significantly elevate accuracy and versatility, as evidenced by empirical data analysis and case studies. The discourse culminates in an exploration of the prospective future enhancements and the expansive applications of this hybrid system in autonomous robotics, highlighting its pivotal role in augmenting operational efficacy and scope [45].

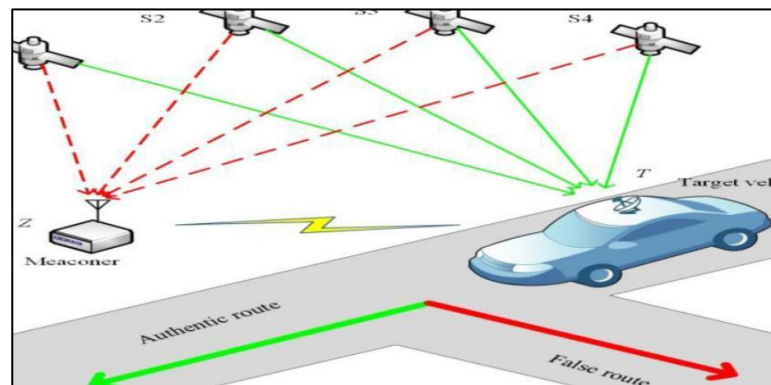


Figure 7 GNSS working principle [45].

2.3.2 Automatic travelling of agricultural support robot for a fruit farm Verification of effectiveness of real-time kinematic-global navigation satellite system and developed a simulator for specification design.

The essay reviews a study by Hiraoka et al. on integrating Real-Time Kinematic-Global Navigation Satellite System (RTK-GNSS) technology into agricultural robots for fruit farms (see figure 8). It highlights the need for such technology due to labor shortages and safety concerns in agriculture, specifically in Japanese orchards. The study compared the effectiveness of RTK-GNSS against Stand-Alone (SA)-GNSS in real orchard settings, revealing that RTK-GNSS significantly outperforms SA-GNSS in terms of accuracy and stability. This improved precision, demonstrated through both field tests and simulations, indicates that RTK-GNSS is highly beneficial for navigating agricultural robots, ensuring more precise and safe operations between tree rows. The findings suggest that RTK-GNSS can revolutionize fruit farm automation, offering a substantial advancement in agricultural technology and reducing the reliance on manual labor. The essay incorporates specific citations and references to the study by [46]., providing a comprehensive overview of its contributions to agricultural robotics.



Figure 8 Photograph of the harvesting route in each field [46].

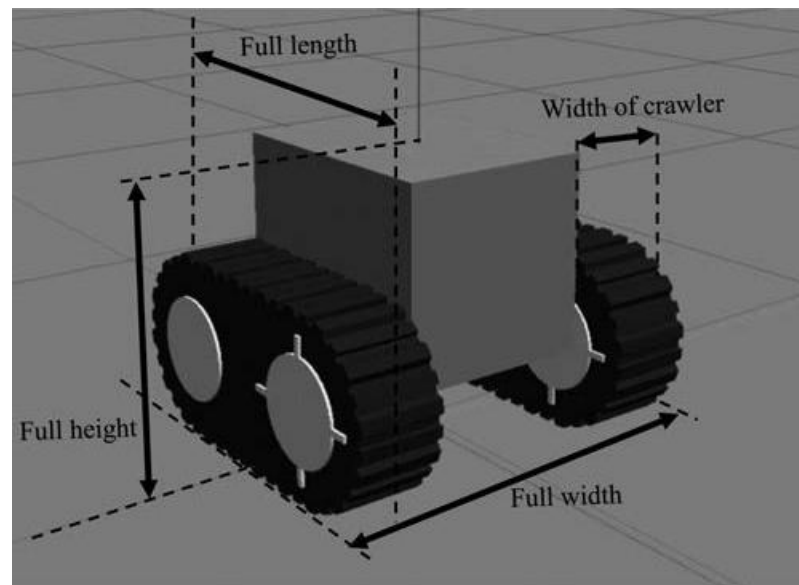


Figure 9 Crawler vehicle model [46].

2.3.2. Autonomous Navigation of an Agricultural Robot Using RTK GPS and Pixhawk

The summary provided already encapsulates the essential points in a formal and concise manner, appropriate for academic or professional contexts. It addresses the significant threat Potato Virus Y (PVY) poses to crops [47], the limitations of current methods for dealing with PVY, and introduces an innovative solution involving Global Positioning System (GPS) guided Autonomous Ground Vehicles (AGVs) with robotic arms. This solution, leveraging advanced remote sensing technologies and Real-Time Kinematic (RTK) systems, aims to improve the accuracy and efficiency of combating PVY spread in potato fields, illustrating a forward-thinking approach to agricultural challenges [48-50].

The integration of an RTK GPS system with magnetometers furnishes the robot with precise positional and directional data. The control logic is managed by a Pixhawk microprocessor, which corrects navigational errors by adjusting the wheel speeds to align the robot's actual position and heading with the intended targets. The choice of the Piksi Multi Evaluation Kit for the RTK GPS, featuring a high-performance Xilinx Zynq 7020 chipset with dual-core ARM Cortex processors, enables rapid RTK calculations, achieving an RTK fix swiftly. This system is highlighted for its cost-effectiveness in comparison to alternative RTK GPS options. Both the base station and rover unit are comprised of four key components, as detailed in the referenced table 3 and figure 10.

Table 3 Piksi multi evaluation kit main parts [33].

Item	Name	Quantity
1	Evaluation board	2
2	Lithium-ion battery	2
3	Free Wave radio modem	2
4	Survey grade GNSS antenna	2

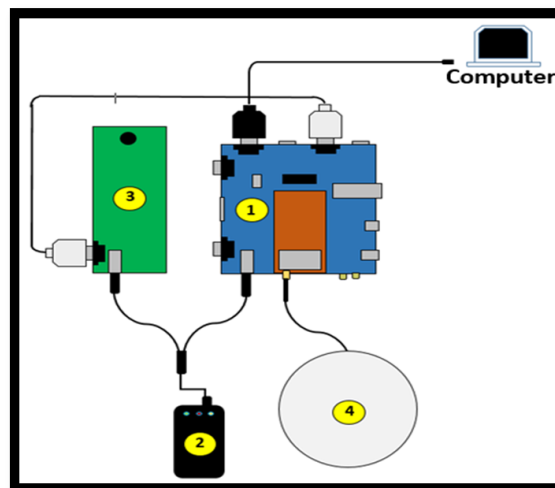


Figure 10 Key components of the Piksi Multi Evaluation Kit for base station and rover setup [50].

Prior to deploying the actual Autonomous Ground Vehicle (AGV), its components underwent preliminary testing utilizing radios supplied by Swift Navigation. The procedure involved configuring the Piksi Multi for RTK mode by programming the base station and rover radios for mutual communication. Subsequently, the components listed in Table 1 were interconnected as depicted in Figure. 11 The setup included connecting a survey-grade GNSS antenna and a Free Wave radio modem to the evaluation board, which was powered alongside the radio modem by a lithium-ion battery. A USB cable was used to connect the evaluation board to a computer. Swift Navigation facilitates this process by offering complimentary software, Swift Console, which aids in configuring and interfacing with the Piksi Multi. For the evaluation phase, radios operating at frequencies of 900 MHz or 2.4 GHz were utilized to assess the GNSS receivers' performance.

The creation of a protective enclosure for the robot's electrical components was crucial to shield them from mud, water, and environmental factors due to its operational setting. Given the prototype nature of the robot, ensuring easy access to these components for maintenance or modifications was a key design consideration. The design process commenced in Solid Works, followed by the fabrication of acrylic sheets through laser cutting and assembly using Weldon #4 acrylic glue (Dichloromethane). Acrylic was chosen for its simplicity in laser cutting, widespread availability, and ability to fit tightly, which improved the protective qualities of the enclosure against environmental factors. The design included precise holes for cable and antenna access, while joints were strengthened with high-temperature resistant hot glue. After assembly, the enclosure was coated in white spray paint to reduce heat absorption from sunlight, thus enhancing the regulation of its internal temperature. (Figure. 11)

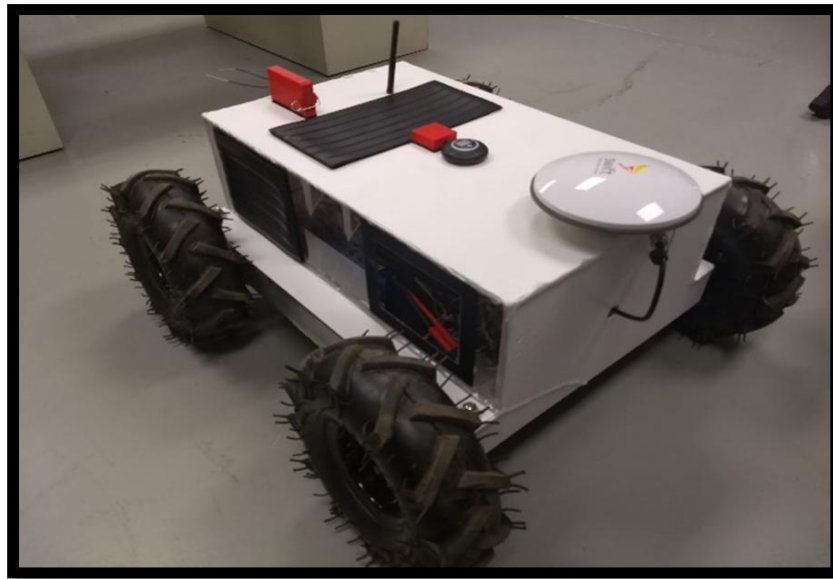


Figure 11 Final assembly of the enclosure mounted on the robot chassis [50].

The enclosure is secured with four bolts, allowing it to be easily removed while keeping the components attached, facilitating simply connect/disconnect operations via access holes. Furthermore, a separate acrylic plate, also laser-cut, serves as the mounting platform for the electrical components, designed for easy removal for maintenance or debugging purposes. The design includes a beneficial aspect for magnetometer calibration, where the bottom plate can stay affixed to the enclosure but be elevated from the chassis, streamlining the calibration procedure.

Piksi Multi Accuracy - The RTK GPS system, specifically the Piksi Multi, demonstrated accuracy within the advertised range of less than or equal to 10 centimeters, verified against National Geodetic Marker sites to an actual precision of 5.6 centimeters. Verification posed challenges due to the scarcity of locations with precisely known GPS coordinates. Two main approaches were evaluated: the direct survey of a location and the utilization of a pre-surveyed geodetic marker. The latter, accessed via the National Geodetic Survey Data Explorer, although more difficult due to access issues on private properties, proved effective. The selected marker was located at Pocatello Regional Airport. During the experiment, evaluation boards from Swift Navigation were utilized, situating the base station's Piksi on one geodetic marker and the rover's Piksi on another, with a distance of several hundred meters between them. After collecting 50 GPS fixes and converting them into the UTM 12N coordinate

system, the average positional error was calculated to be 5.6 centimeters, using Pythagorean's theorem to compare the estimated and actual locations. Testing was constrained by location difficulties and limited time permitted by airport security.

Autonomous Navigation Via Waypoints - The autonomous navigation system performed adequately, although issues were noted with the AGV's ability to maintain its heading, causing occasional disorientation. This issue was considered minor and could likely be resolved by employing better-quality magnetometers or achieving more accurate calibration. While the accuracy of autonomous navigation was not tested against geodetic markers due to prior confirmation, challenges associated with the waypoint radius parameter were identified, suggesting the need for refinement. A proposed solution involves integrating a trigger event for enhanced precision in navigation, especially when approaching targeted waypoints within a 1-meter radius.

Obstacle Avoidance - In the experiment, SiK telemetry radios were replaced with a direct USB connection. This setup allowed for accurate distance measurements between the LIDAR and a frontal obstacle on the AGV, aligning with the readings in Mission Planner. However, obstacle avoidance capabilities were not tested using the telemetry connection. Future evaluations are planned to incorporate telemetry connections for comprehensive testing.

CHAPTER III

RESEARCH METHODOLOGY

Materials

1. Hardware
 - Personal computer
 - Mobile robot
2. Software
 - Robot Operating System2 (ROS2)
 - Visual Studio Code

Methodology

The present study is dedicated to the engineering of a navigation system designed for robotic deployment in extensive agricultural landscapes. This includes an empirical demonstration of the robots' navigational capabilities. The research methodology is delineated in a sequential and structured manner, partitioned into four main sections:

- 3.1 Robot Vehicle Design
- 3.2 Navigation System Design of the robot
- 3.3 Software design
- 3.4 GNSS-IMU fusion Position Principle
- 3.5 IMU-Based Heading Angle Measurement
- 3.6 Trajectory-Tracking Control System
- 3.7 Development and Implementation of Algorithms
- 3.8 Implementation procedure

3.1 Robot Vehicle Design

3.1.1 Hardware Integration

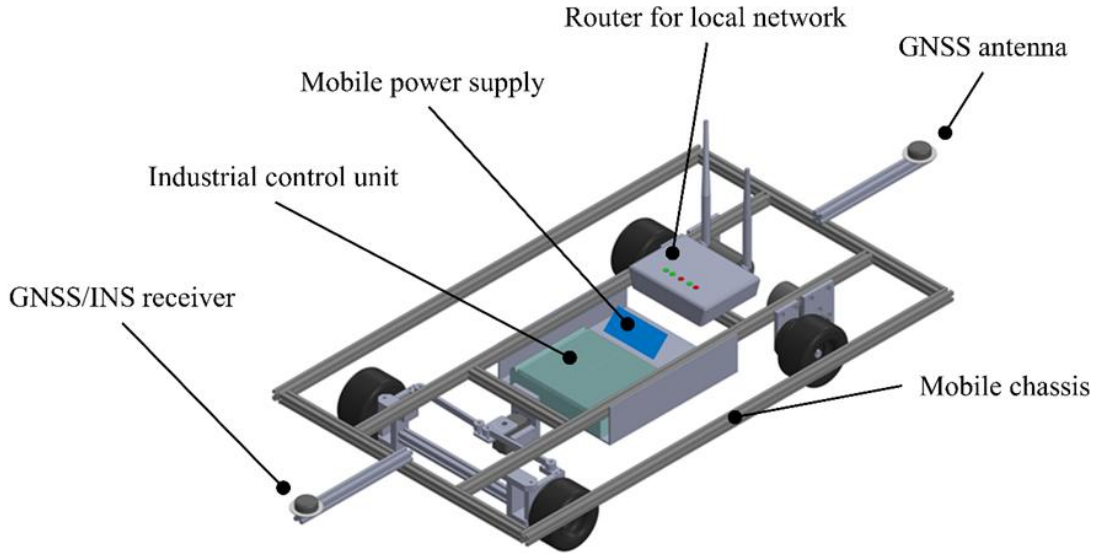


Figure 12 Hardware components of robotic vehicles.

This study employs a mobile robot featuring an aluminum profile chassis, an industrial-grade onboard computer, GNSS antennas, a high-precision receiver, and brushless DC motors. The chassis of the robot measures 600 mm in width and 1000 mm in length. The GNSS system incorporates an internal IMU and supports RTK positioning with an accuracy of up to 1.5 centimeters, delivering complete differential data compliant with the Radio Technical Commission for Maritime Services (RTCM) standards. Driven by a high-performance MEMS-based inertial device, the GNSS-RTK module ensures precise and reliable differential positioning. Additionally, the robot is equipped with a local wireless communication system that facilitates real-time control between the ground station and the mobile platform. This communication system supports full-duplex transmission, real-time channel monitoring, and adjustable transmission power, thereby ensuring reliable operation under diverse outdoor conditions. The comprehensive hardware architecture of the robot is depicted in Figure 12 and summarized in Table 4.

Table 4 Main parameter of robot.

Parameter	Unit	Value
Dimensions	mm	600*1000*110
Dimensions + Antenna	mm	600*1480*110
Wheel	mm	113
Wheelbase	mm	613.5
Max speed	m/s	1.0-5.0
Power supply	volt	24

The electrical system of the autonomous mobile robot is systematically organized to ensure stable and efficient operation. Power is initially supplied by a 24VDC battery, which is regulated and distributed to various subsystems through voltage converters. These converters maintain consistent voltage levels, providing stable power to critical modules.

The motor controllers receive the regulated power and are responsible for managing the speed and direction of the robot's brushless DC motors. Simultaneously, the IMU and GNSS-RTK modules, also powered through the regulated supply, continuously acquire real-time navigation data to support precise localization and path tracking.

In parallel, the wireless communication modules utilize the same power infrastructure to establish a real-time connection between the robot and the ground station, enabling remote control and seamless data exchange.

The overall structure of the electrical system, detailing the power flow from the battery through each subsystem and the interconnection between components, is illustrated in Figure 13.

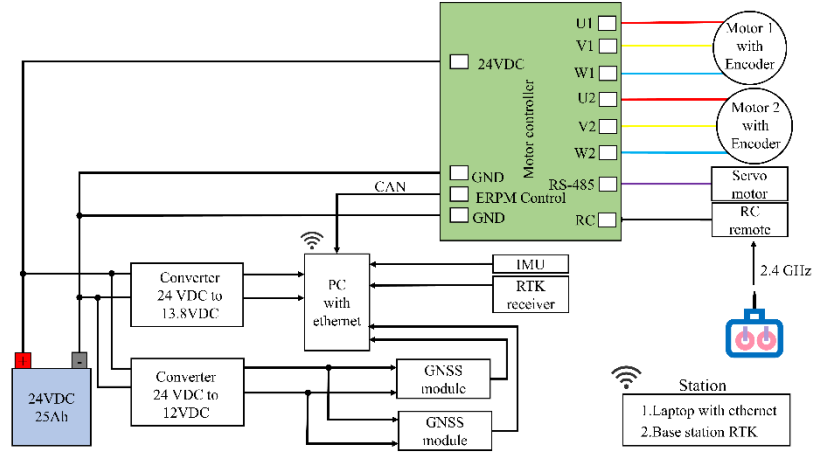


Figure 13 Power and communication layout of the AMR.

3.2 Navigation System Design of the robot

The robot's navigation system is designed using a high-precision architecture centered on GNSS-RTK technology to ensure accurate and reliable localization in outdoor environments. As depicted in Figure 4, the system is structured around four principal components: the GNSS-RTK base station, the position unit, the processing unit, and the control unit.

The GNSS-RTK base station, comprising a GNSS antenna and a wireless transmission module, serves as the source of differential correction data. This correction data is transmitted to the position unit, which consists of a wireless receiver and a GNSS module. By combining satellite signals with the received corrections, the position unit achieves centimeter-level positioning accuracy.

The processing unit, based on an industrial-grade computer, fuses data from the GNSS and an onboard IMU to enhance localization performance, particularly in areas with limited satellite visibility. It transmits motion commands to the control unit via the Controller Area Network (CAN) bus protocol, ensuring low-latency and reliable real-time communication.

The control unit, which includes brushless DC (BLDC) motors, motor controllers, and encoder sensors, executes the received commands and provides real-time feedback to support closed-loop control of the robot's motion.

This integrated system design enables precise, stable, and responsive navigation, making it highly suitable for applications such as autonomous agricultural machinery

and field robotics. Furthermore, the adoption of standardized communication protocols and modular hardware components enhances the system's reliability, scalability, and adaptability across diverse operational scenarios [51–52].

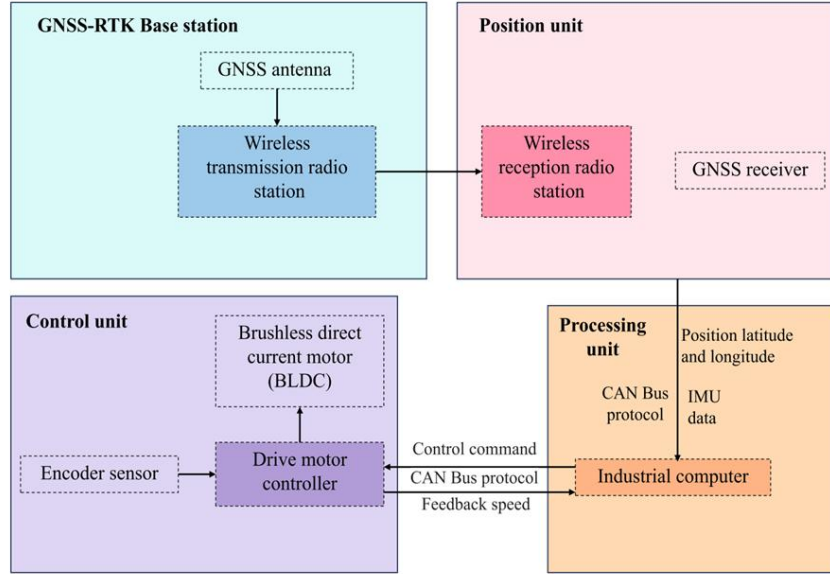


Figure 14 Integrated GNSS-RTK-based navigation system.

The software system responsible for controlling the autonomous mobile platform is implemented on an embedded computing platform operating under the Linux operating system. It employs the Robot Operating System (ROS) as the core integration middleware for managing autonomous positioning and navigation, as depicted in Figure 16. This platform continuously acquires data from multiple onboard sensors, including a Global Navigation Satellite System (GNSS) receiver, an Inertial Measurement Unit (IMU), and wheel encoders. Together, these sensors provide critical information regarding the robot's positioning, orientation, and motion, which are essential for real-time localization and motion control.

The robot is equipped with satellite-based self-positioning sensors (GNSS) and an IMU, both of which continuously publish data streams within the ROS ecosystem. The IMU plays a crucial role in determining the robot's heading by providing orientation data necessary for maintaining accurate directional control, while the wheel encoders supplement motion information by reporting linear and angular velocities. All sensor data streams are collected, processed, and integrated by a sensor fusion module,

which performs real-time analysis to ensure reliable localization and navigation performance [11–12].

Processing and fusing the sensor data, the software system generates motion control commands aligned with the navigation objectives. These commands are transmitted to the motion control module, responsible for activating the drive system, which includes the left and right brushless DC motors along with the steering mechanism. Following receipt of the commands, the drive controller initiates movement by adjusting the steering and motor operations, guiding the robot along the desired trajectory.

The architecture of the software system is meticulously designed to ensure seamless coordination among sensing, processing, and actuation components. Leveraging the flexibility and robustness of Linux and ROS, the system distributes its functionalities across multiple ROS nodes dedicated to sensor data acquisition, control signal generation, and inter-component communication.

As illustrated in Figure 15, a planning module is utilized to generate the target path, which is subsequently tracked by the Pure Pursuit controller. The integration of GNSS and IMU data enables the robot to navigate with high accuracy, allowing it to adapt effectively to varying environmental conditions. The comprehensive system architecture is depicted in both Figure 15 and Figure 16, demonstrating the interactions among system modules that collectively support autonomous and precise robot operation [13–14].

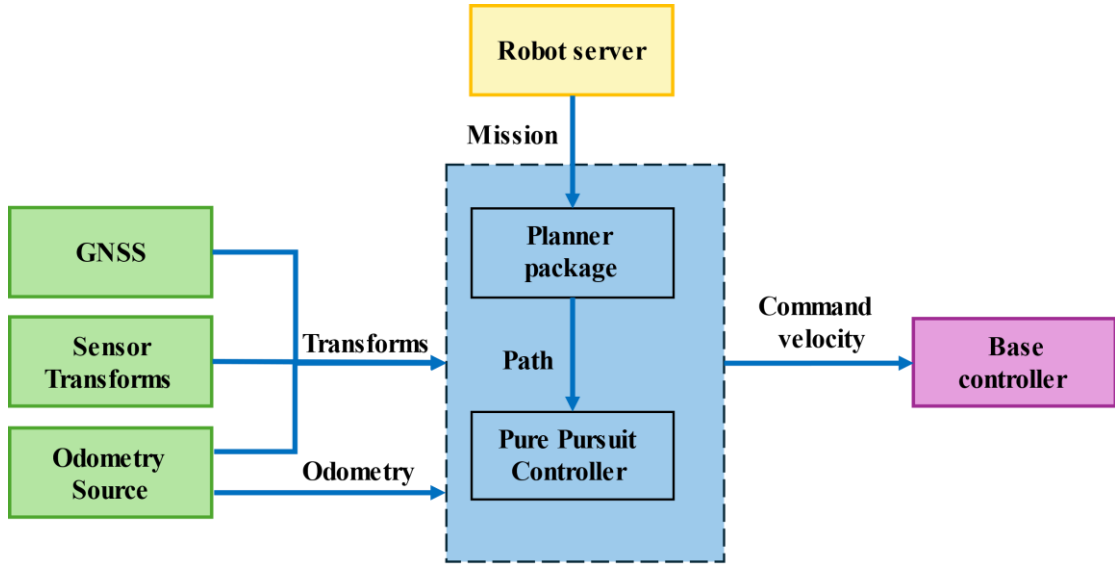


Figure 15 Functional flow of GNSS-based navigation using a PP controller.

3.3 Software design

3.3.1 Robot Operating System2 (ROS2)

Robot Operating System 2 (ROS 2) Jazzy is a modern robotics development framework that builds upon the foundation established by ROS 1, introducing major advancements in communication architecture, real-time capabilities, modularity, and security. In the domain of mobile agricultural robotics, ROS 2 Jazzy significantly enhances the development of autonomous systems by offering robust support for multi-node, distributed, and scalable communication architectures. This capability is enabled through its adoption of the Data Distribution Service (DDS) middleware, which ensures low-latency and high-reliability message delivery. Such performance is particularly important in dynamic environments, such as open agricultural fields.

In addition, ROS 2 Jazzy integrates more effectively with simulation tools like Gazebo and provides enhanced visualization capabilities through RViz2. These tools are highly beneficial for testing navigation algorithms, including Pure Pursuit. Recent improvements in ROS 2 Jazzy also include the integration of Zenoh-based middleware, which is optimized for low-bandwidth networks, and enhancements in handling large ROSBag datasets. These features facilitate efficient recording and replaying of robot state information during field trials.

With these advancements, ROS 2 Jazzy is particularly well-suited for applications that demand real-time responsiveness, system flexibility, and modular design in challenging outdoor environments. The literature emphasizes that ROS 2 Jazzy represents a major milestone in the evolution of the ROS ecosystem. It successfully addresses previous limitations in scalability, security, and robustness, thus enabling more reliable and resilient autonomous operations under demanding field conditions [22].

3.3.2 Comprehensive Development Framework

Visual Studio Code (VS Code) version 1.75, developed by Microsoft, is a lightweight yet versatile source code editor that supports multiple programming languages, including C, C++, and Python. It offers features such as intelligent code completion, an integrated debugger, and built-in version control, which collectively enhance the efficiency of software development processes.

Its seamless integration with Python 3 facilitates rapid implementation and testing of algorithms, thereby improving both computational accuracy and overall workflow productivity. Due to its flexibility and extensibility, VS Code is well-suited for developing and maintaining complex algorithmic systems in research and engineering applications.

3.4 GNSS-RTK and IMU fusion Position Principle

3.4.1 Global Navigation Satellite System Real-Time Kinematic (GNSS RTK)

Global Navigation Satellite System (GNSS) and Real-Time Kinematic (RTK) positioning technologies have significantly advanced the field of precision agriculture, particularly in mobile robotic systems. GNSS-RTK enables centimeter-level positioning accuracy, which is essential for autonomous operations such as seeding, fertilizing, and mapping agricultural fields. This level of precision reduces redundancy, minimizes input waste, and enhances resource efficiency and yield outcomes. Moreover, GNSS-RTK enhances the reliability of robotic operations under varying environmental conditions, including changes in weather and lighting, thus improving overall system productivity.

To apply GNSS data within a Cartesian coordinate system suitable for robotic navigation, it is necessary to convert geodetic coordinates (latitude and longitude) into the Universal Transverse Mercator (UTM) coordinate system. The UTM system

provides a two-dimensional, metric-based map projection of the Earth's surface, simplifying calculations related to distance and direction, which are critical for accurate motion control[15].

In UTM coordinates, a location is specified using a zone number, hemisphere, and a pair of easting and northing values. Each zone spans 6° of longitude, with a central meridian assigned and a false easting of 500,000 meters to prevent negative coordinate values. For example, an easting of 400,000 meters corresponds to a position roughly 100 kilometers west of the central meridian. In the northern hemisphere, northing values start at 0 at the equator and increase up to approximately 9.3 million meters at 84°N, whereas in the southern hemisphere, northings begin at 10 million meters and decrease southward[22].

The coordinate transformation process is based on the World Geodetic System 1984 (WGS 84), which models the Earth as an oblate spheroid with a semi-major axis of $a = 6,378.137$ meters and an inverse flattening of $\frac{1}{f} = 298.25722356$. The globe is divided into 60 longitudinal zones, each with its own central meridian (λ_0).

The transformation from geodetic coordinates to UTM involves several computational steps. The UTM easting (E) and northing (N) are calculated using the following equations, which include a false easting (E^0), false northing (N^0), and a central scale factor (k^0), typically set to 0.9996. Intermediate values ξ' and η' are derived from the trigonometric transformations of latitude (ϕ) and the difference between the longitude (λ) and the central meridian.

To convert from geodetic coordinates (ϕ, λ) to UTM coordinates (E, N), the transformation given in Eqs. (1) and (2):

$$E = E_0 + k_0 \cdot A \cdot (\eta' + \sum_{j=1}^3 \alpha_j \cos(2j\xi') \sinh(2j\eta')), \quad (1)$$

$$N = N_0 + k_0 \cdot A \cdot (\xi' + \sum_{j=1}^3 \alpha_j \sin(2j\xi') \cosh(2j\eta')), \quad (2)$$

In these Eqs. (1) and (2), E and N are the resulting UTM easting and northing coordinates, respectively. The terms E_0 and N_0 represent the false easting and false northing used to shift the origin of the UTM grid. The scaling constant k_0 is the central scale factor (typically 0.9996), used to minimize distortion along the central meridian. The variable A is a scaling parameter dependent on the ellipsoid geometry, and α_j are coefficients derived from a Fourier series expansion related to the shape of the ellipsoid.

The terms ξ' and η' , required in Eqs. (1) and (2), are intermediate coordinates computed from geodetic latitude and longitude, as defined in Eqs. (3) and (4):

$$\xi' = \tan^{-1} \left(\frac{t}{\cos(\lambda - \lambda_0)} \right), \quad (3)$$

$$\eta' = \tanh^{-1} \left(\frac{\sin(\lambda - \lambda_0)}{\sqrt{1 + t^2}} \right), \quad (4)$$

Here, $t = \tan(\phi)$ represents the tangent of the geodetic latitude, and λ_0 is the longitude of the central meridian for the corresponding UTM zone. The function $\tanh^{-1}(\phi)$ denotes the inverse hyperbolic tangent, and the use of trigonometric and hyperbolic functions accounts for the Earth's curvature during projection.

The constants required for these calculations include the eccentricity e of the ellipsoid, which is derived from its semi-major axis a and semi-minor axis b , as shown in Eq. (5):

$$e^2 = 1 - \frac{b^2}{a^2}, \quad (5)$$

$$A = \frac{a}{(1 - e^2)^{1/4}}, \quad (6)$$

Where e^2 represents the squared eccentricity of the ellipsoid, quantifying its deviation from a perfect sphere. The semi-minor axis b can be obtained from the inverse flattening factor. These definitions ensure precise modeling of Earth's geometry for accurate coordinate transformation.

By applying for Eqs. (1) – (6), GNSS geodetic data can be accurately transformed into metric UTM coordinates suitable for robotic applications. This transformation forms the foundation for fusing GNSS data with inertial measurements from an IMU, allowing for robust and continuous localization even in areas where satellite visibility is compromised.

3.5 IMU-Based Heading Angle Measurement

The calculation of the rotation or heading of mobile robots is a critical aspect of autonomous navigation and robotic system design. Accurate determination of a robot's orientation is essential for executing tasks such as path planning, obstacle avoidance, and precise maneuvering within dynamic environments. Inertial Measurement Units (IMUs) play a fundamental role in this process. These sensors, which typically consist of gyroscopes and accelerometers, are employed to measure both rotational movements and linear accelerations of mobile robots. The gyroscope measures the rate of rotation

around the robot's axes, while the accelerometer captures linear acceleration along these axes. By integrating the data from these two types of sensors, the robot's orientation over time can be accurately calculated.

The heading angle $\theta(t)$ at a given time can be estimated by integrating the angular velocity $\omega(t)$ measured by the gyroscope. The basic equation for updating the heading is expressed Eq. (7):

$$\theta(t) = \theta(t - 1) + \omega(t) \cdot \Delta t, \quad (7)$$

Here, $\theta(t)$ is the current heading angle, $\theta(t - 1)$ is the previous heading, $\omega(t)$ is the angular velocity at time t , and Δt is the time interval between measurements [53]. By continuously applying Eq. (7), the robot's heading can be tracked over time. An illustration of the robot's heading and rotational motion is shown in Figure 16.

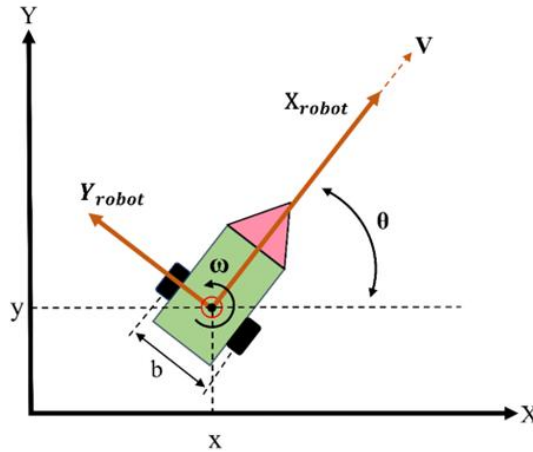


Figure 16 Mobile robot odometry model.

3.6 Trajectory-Tracking Control System

3.6.1 Car-Like Kinematic Model of the Robotic Vehicle

The car-like kinematic model provides a fundamental framework for representing the motion of four-wheeled, steering-type mobile robots, particularly those with automobile-like configurations. This model is extensively utilized in autonomous navigation systems because of its simplicity and its ability to accurately predict motion trajectories while adhering to nonholonomic constraints [54–55]. It serves as the foundational basis for many trajectories tracking algorithms, including the widely used Pure Pursuit method, where the geometric relationships between steering angles,

vehicle velocities, and curvature radii are critical for calculating the target point and generating appropriate control commands.

In the context of Pure Pursuit, the car-like kinematic model enables the robot to dynamically adjust its steering based on the relative position of a lookahead point along the planned path. By leveraging the mathematical relationship between the curvature of the desired path and the required steering angle, the robot is able to continuously update its heading and maintain smooth and accurate path following. The simplicity and efficiency of this model make it particularly well-suited for real-time applications in outdoor environments, where fast response and adaptability to path deviations are essential.

Furthermore, because the car-like kinematic model captures the nonholonomic motion constraints inherent to wheeled robots, it provides a realistic and reliable foundation for designing navigation and control strategies. These characteristics are especially important in agricultural, field robotics, and other outdoor scenarios, where unpredictable terrain and environmental disturbances necessitate robust and precise control mechanisms to ensure stable and effective autonomous operation.

As illustrated in Figure 17, the vehicle's configuration is defined with respect to the rear axle center (x, y) , its heading angle (θ) , and the front-wheel steering angle (ϕ) . The wheelbase (L) refers to the distance between the front and rear axles. The forward velocity is denoted (v_1) , and the steering rate is represented by (v_2) , which collectively influences the robot's motion over time. The position of the front axle (x_f, y_f) is derived geometrically from the rear axle, and a designated target point (P) on the reference trajectory is used for tracking control. The distance between the front axle and this target point is denoted (R) , which acts as the dynamic lookahead parameter in trajectory-following algorithms.

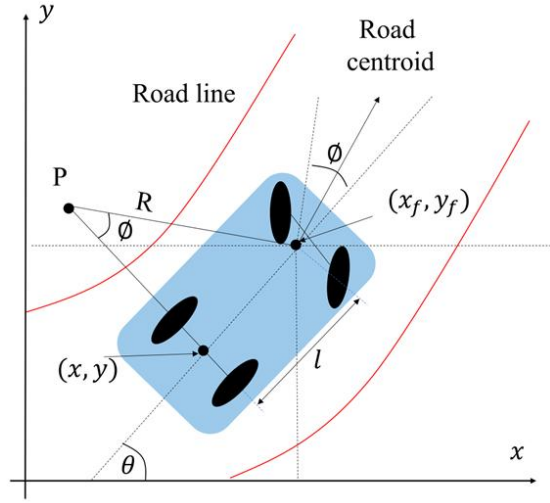


Figure 17 Kinematic structure of a car-like vehicle.

In discrete time, the robot's motion is governed by Eqs. (8) and (9), which update the x and y coordinates based on the current velocity and heading angle. The orientation update is derived from Eq. (10), which relates the turning curvature to the steering angle:

$$x(t+1) = x(t) + v(t) \cdot \cos(\theta(t)) \cdot \Delta t, \quad (8)$$

$$y(t+1) = y(t) + v(t) \cdot \sin(\theta(t)) \cdot \Delta t, \quad (9)$$

$$\theta(t+1) = \theta(t) + \frac{v(t)}{L} \cdot \tan(\delta(t)) \cdot \Delta t, \quad (10)$$

To support real-time control and simulation, the model can also be reformulated in continuous-time state-space form as shown in Eq. (11), where the forward velocity v_1 and steering rate v_2 serve as control inputs:

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \\ \dot{\phi} \end{bmatrix} = \begin{bmatrix} \cos \theta \\ \sin \theta \\ \frac{\tan \theta}{L} \\ 0 \end{bmatrix} v_1 + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} v_2 \quad (11)$$

Here, v_1 represents the forward linear velocity, while v_2 corresponds to the steering rate. The time step Δt governs the resolution of discrete-time updates, enabling real-time implementation on robotic platforms.

3.6.2 Pure Pursuit algorithm

The Pure Pursuit (PP) algorithm is a geometric-based path-tracking method that is widely utilized in mobile robotic systems because of its simplicity and effectiveness. The fundamental principle of the algorithm is to identify a goal point

located ahead along the reference trajectory and to compute a circular arc that connects the vehicle's current position to this target point. Based on this arc, the algorithm determines the appropriate front-wheel steering angle required for the vehicle to follow the desired path. As illustrated in Figure 18, the center of the vehicle's rear axle, denoted as (x, y) , is used as the reference point for motion control, while the lookahead point is represented by (x_l, y_l) . A critical parameter in the Pure Pursuit algorithm is the lookahead distance Ld , which defines how far ahead along the path the goal point is selected. The value of Ld has a significant influence on the robot's steering behavior and trajectory smoothness. A shorter lookahead distance results in more aggressive steering commands, allowing the robot to respond quickly to path deviations but potentially causing oscillations and instability at higher speeds. Conversely, a longer lookahead distance generates smoother steering trajectories and improved stability, although it may reduce the system's ability to track sharp turns accurately. Therefore, selecting an appropriate Ld is crucial for balancing responsiveness and stability, and in some implementations, Ld is dynamically adjusted based on the vehicle's velocity and operating conditions to optimize path-following performance.

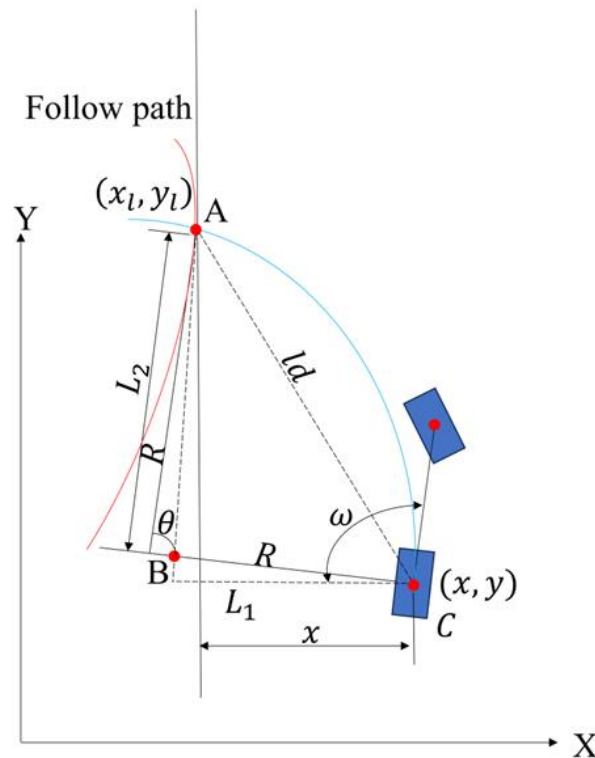


Figure 18 Geometric representation of the PP path tracking.

The Euclidean distance between the current vehicle position and the target point defines the lookahead distance ld , as formulated in Eq. (12):

$$l_d = \sqrt{(x - x_l)^2 + (y - y_l)^2}, \quad (12)$$

From the geometric relationships in the figure, the intermediate distances L_1 and L_2 , which connect auxiliary points B , C , and A , are derived using the radius of the curvature R and turning angle θ , as shown in Eq. (13):

$$\begin{cases} L_1 = -R(1 - \cos(\theta)) \\ L_2 = -R \sin(\theta) \end{cases}, \quad (13)$$

Using the Pythagorean identity, the relationship between these distances yields:

$$L_1^2 + L_2^2 = l_d^2, \quad (14)$$

In practice, the value of L_1 can be alternatively expressed based on the vehicle's current lateral offset and steering angle ω as shown in Eq. (15):

$$L_1 = -x \cos\left(\omega - \frac{\pi}{2}\right) + \sqrt{l_d^2 - x^2} \sin\left(\omega - \frac{\pi}{2}\right), \quad (15)$$

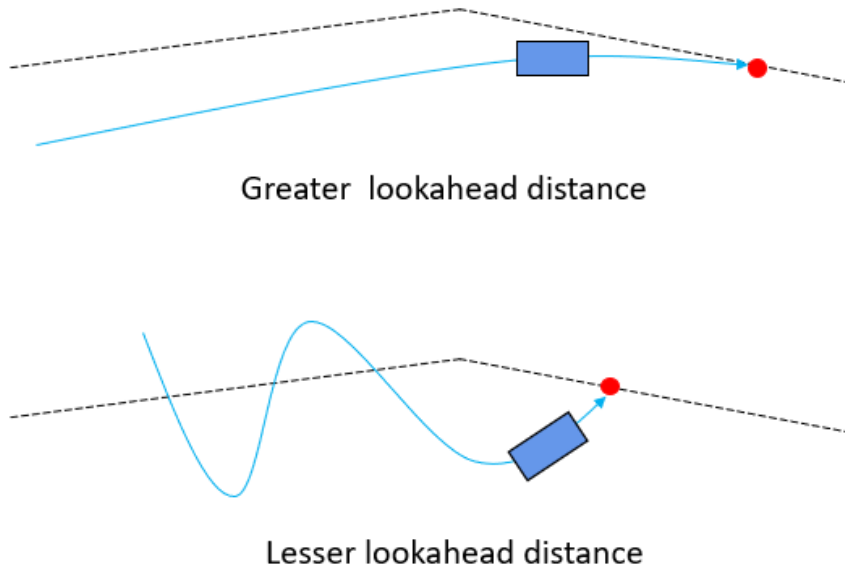


Figure 19 Diagram illustrates the effects of varying look-ahead distances.

The instantaneous turning radius R is directly related to the steering angle δ_U by Eq. (16):

$$R = \frac{L}{\tan(\delta_U)}, \quad (16)$$

Substituting Eqs. (15) and (16) into Eq. (14) and rearranging, we obtain:

$$2R \times L_1 = l_d^2, \quad (17)$$

Solving the δ_U , the final expression is given in Eq. (18):

$$\delta_U = \arctan \left(\frac{2L \left(x \sin \phi + \sqrt{l_d^2 - x^2 \cos \phi} \right)}{l_d^2} \right), \quad (18)$$

From this, the path curvature γ_U can be calculated using Eq. (19):

$$\gamma_U = \frac{\tan \delta_U}{L} = \frac{2 \left(x \sin \phi + \sqrt{l_d^2 - x^2 \cos \phi} \right)}{l_d^2}, \quad (19)$$

As indicated by Eqs. (18) and (19), the lookahead distance ld plays a critical role in regulating both the steering angle δ_u and the turning curvature γ_u . Larger values of ld result in smaller steering angles, leading to smoother but slower directional corrections. In contrast, smaller values cause rapid, aggressive adjustments that may induce oscillatory behavior, especially during sharp turns or dynamic maneuvers, as illustrated in Figure 19. This phenomenon has been similarly reported in previous studies [56–57], which emphasize the importance of appropriately tuning the lookahead parameter to balance responsiveness and stability.

Selecting an optimal ld is particularly crucial in high-speed navigation tasks, where maintaining both trajectory accuracy and motion stability becomes increasingly challenging. Adaptive strategies that adjust the lookahead distance based on vehicle velocity and steering demands have been proposed to further enhance path-following performance under varying environmental and operational conditions.

3.6.3 Path tracking analysis

An analysis of path tracking strategies with an emphasis on enhancing the PP algorithm through a dynamic lookahead distance mechanism. In its traditional form, the PP algorithm utilizes a fixed lookahead distance ld , which may lead to suboptimal tracking performance under varying velocity conditions. A small constant ld often results in oscillatory behavior due to overly aggressive steering, whereas a large ld can delay corrective actions and degrade path-following accuracy.

To improve adaptability, the lookahead distance can be dynamically adjusted in response to the robot's current linear velocity. This allows the controller to assign a shorter ld at low speeds for precise maneuvering and a longer ld at higher speeds to ensure smooth and stable motion. Figure 8 illustrates this relationship by comparing the

qualitative effects of different lookahead distances. A greater ld results in smoother trajectory tracking but reduces responsiveness, while a smaller ld enables sharper corrections that may cause instability or oscillations, particularly during low-speed maneuvers.

The dynamic lookahead distance is defined as a piecewise function of the absolute linear velocity v , as shown in Eq. (20), where v represents the robot's current forward speed in meters per second:

$$ld = \begin{cases} 1.5 + (|v| - 1.0) \times 3.0 & \text{if } (|v| - 1.0) \times 3.0 \leq 10.0 \\ 1.5 + 10.0 & \text{otherwise} \end{cases}, \quad (20)$$

In this formulation, 1.5 meters serves as the base lookahead distance. The dynamic component increases linearly with the velocity offset from 1.0 m/s, scaled by a factor of 3. However, to avoid oversteering at high speeds, the maximum lookahead distance is limited to 11.5 meters.

As part of the geometric derivation, the variable ω represents the angular offset between the robot's heading direction and the line toward the goal point. The steering angle required to reach the target is denoted by δ_U , which is calculated using the dynamic lookahead distance in Eq. (18). Subsequently, the path curvature resulting from this steering command is represented by γ_U , as shown in Eq. (19). These variables allow the controller to adaptively compute turning behavior according to real-time speed and position.

By applying Eq. (20), the PP controller becomes more responsive and stable across varying speeds. This approach is particularly beneficial in agricultural and field robotics, where terrain and motion dynamics can change rapidly.

3.7 Development and Implementation of Algorithms

3.7.1 Pure Pursuit Path Tracking (PP)

Pure Pursuit calculates the steering angle δ and linear velocity v required to follow a predefined path. It identifies a goal point located a fixed distance ld ahead of the robot's position (x, y) , and computes a circular arc toward it based on the θ and L .

The reference path is represented as a discrete set of waypoints $P = \{(x_i, y_i)\}_{i=1}^n$. The algorithm finds the nearest waypoint, then selects the first point beyond the lookahead distance. This point is transformed into the robot's local

coordinate frame and used to calculate δ . The velocity v is assigned either as a constant or adaptive value. The full procedure is shown in Algorithm 1.

Algorithm 1: Pure Pursuit Path Tracking (PP)

Input: Current robot state (x, y, θ) , path (P) , wheelbase (L) , fixed lookahead distance (ld)

Output: Steering angle δ , linear velocity v

1. Acquire the current position of the robot: (x_{robot}, y_{robot}) .
2. Acquire the current heading angle of the robot: θ_{robot} .
3. Find the index of the nearest waypoint on the path:
 $nearest_{idx} \leftarrow \text{Find closest waypoint } (P, x_{robot}, y_{robot})$.
4. Starting from $nearest_{idx}$, search for the lookahead point:
 - For each point $P[i]$ from $nearest_{idx}$ to the end of the path:
 - If the distance between $P[i]$ and the current position is greater than or equal to ld , set
 $lookahead_{point} \leftarrow P[i]$ and break the loop.
5. Transform the coordinates of the $lookahead_{point}$ into the robot's local (base_link) frame:
 $(x_{ld}, y_{ld}) \leftarrow \text{TransformToRobotFrame}(lookahead_{point}, x_{robot}, y_{robot}, \theta_{robot})$
6. Calculate the steering angle using the Pure Pursuit formula:
 $\delta \leftarrow \text{atan2}(2 \cdot y_{ld} \cdot L, ld^2)$.
7. Constrain the steering angle within the maximum allowable limits:
 $\delta \leftarrow \text{constrain}(\delta, -\delta_{max}, \delta_{max})$.
8. Set the linear velocity within predefined safe bounds:
 $v \leftarrow \text{constrain}(v_{desired}, v_{min}, v_{max})$.
9. Return the computed steering angle δ and linear velocity v .

Although this implementation uses a fixed lookahead distance, the algorithm structure supports further enhancement, such as dynamic lookahead control.

3.7.2 Pure Pursuit Path Tracking with Dynamic Lookahead Distance (PP-DL)

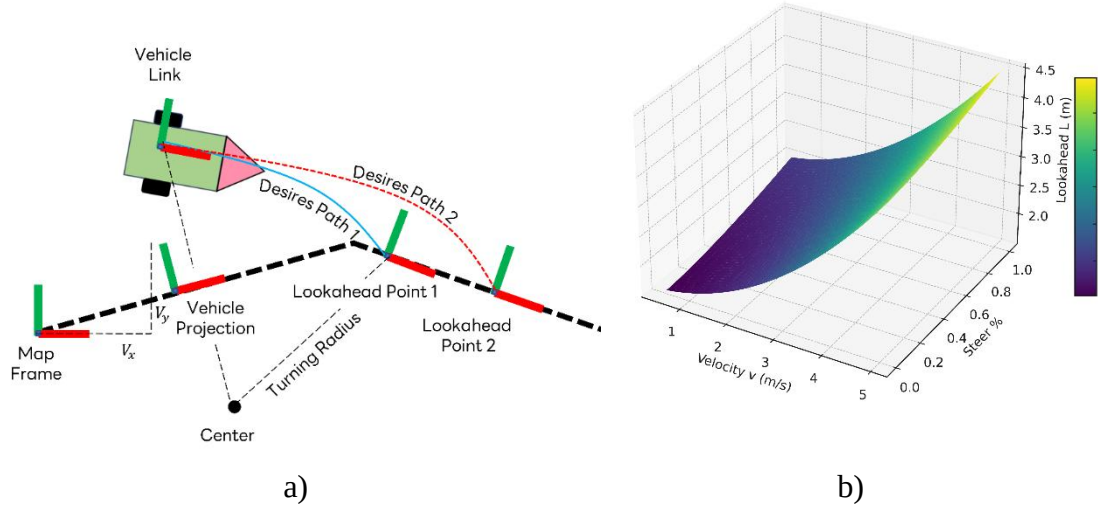


Figure 20 Pure pursuit tracking model and adaptive lookahead surface: (a) Geometric representation of lookahead-based path tracking; (b) Lookahead distance as a function of velocity and steering ratio.

To enhance tracking performance across varying speeds and motion contexts, the traditional PP algorithm is extended by introducing a dynamic lookahead distance, ld , which adjusts in real time based on the robot's velocity. This technique improves the accuracy of tight turns at low speeds and provides smooth trajectory following at higher speeds.

As depicted in Figure 20(a), the vehicle determines its steering angle by computing a circular arc connecting its current rear-axle position to a lookahead point on the desired path. The steering angle δ is calculated using the standard geometric formulation shown in Eq. (21):

$$\delta = \tan^{-1} \left(\frac{2y_{ld}L}{ld^2} \right), \quad (21)$$

To prevent physically unrealistic steering angles, a saturation condition is enforced using the piecewise constraint shown in Eq. (22):

$$\delta = \max(\min(\delta, \delta_{max}), -\delta_{max}), \quad (22)$$

The lookahead distance ld is linearly adjusted based on the robot's current velocity, bounded by minimum and maximum limits. The adaptive formulation is expressed in Eq. (23):

$$ld = ld_{min} + \left(\frac{v - v_{min}}{v_{max} - v_{min}} \right) (ld_{max} - ld_{min}), \quad (23)$$

Velocity constraints for smooth control are expressed in Eqs. (24) and (25):

$$v_{accelerate} = v_{robot} + accelerate \times \Delta t, \quad (24)$$

while the velocity constrained by deceleration is computed as Eq. (25):

$$v_{decelerate} = \sqrt{2 \times decelerate \times \text{remaining distance}}, \quad (25)$$

The final velocity command is selected using Eq. (26):

$$v = \begin{cases} v_{max} & \text{if } \min(v_{acc}, v_{dec}) > v_{max} \\ v_{min} & \text{if } \min(v_{acc}, v_{dec}) < v_{min} \\ \min(v_{acc}, v_{dec}), & \text{otherwise} \end{cases} \quad (26)$$

As shown in Figure 20(b), the lookahead distance ld increases proportionally with the robot's forward velocity, ranging from ld_{min} to ld_{max} . This adaptive scaling enables tighter turning at low speeds and smoother tracking at higher speeds. By combining Eqs. (21) through (26), the controller jointly adjusts both the steering angle and linear velocity based on the robot's state and motion constraints, enhancing robustness and trajectory stability across varying operating conditions.

Algorithm 2: Pure Pursuit Path Tracking with Dynamic Lookahead Distance (PP-DL)

Input Current robot state (x, y, θ) , path (P) , wheelbase (L) , lookahead limits

(ld_{min}, ld_{max}) , velocity limits (v_{min}, v_{max}) .

Output: Steering angle δ , linear velocity v .

1. Retrieve the current robot state, including position, heading, and velocity:

$$(x_{robot}, y_{robot}, \theta_{robot}, v_{robot}).$$

2. Find the index of the nearest waypoint on the path:

$$nearest_{idx} \leftarrow \operatorname{argmin}_i \text{distance}(P[i], (x_{robot}, y_{robot})).$$

3. Compute the normalized velocity ratio:

$$velocity_ratio \leftarrow \frac{v_{robot} - v_{min}}{v_{max} - v_{min}}.$$

4. Clamp the velocity ratio to the range $[0, 1]$:

$$velocity_ratio \leftarrow \text{clamp}(velocity_ratio, 0, 1).$$

5. Calculate the dynamic lookahead distance:

$$ld \leftarrow ld_{min} + velocity_{ratio} \times (ld_{max} - ld_{min}).$$

6. Search for the lookahead point on the path:

For each point $P[i]$ starting from $nearest_idx$,

if $distance(P[i], (x_{robot}, y_{robot})) \geq ld$ then assign

$lookahead_point \leftarrow P[i]$ and break.

7. Transform the lookahead point into the robot's local ($base_link$) frame:

$(x_{ld}, y_{ld}) \leftarrow \text{Transform to robot frame}$

$(lookahead_point, x_{robot}, y_{robot}, \theta_{robot})$.

8. Calculate the steering angle using the Pure Pursuit formula:

$$\delta \leftarrow atan2(2 \cdot y_{ld} \cdot L, ld^2).$$

9. Constrain the steering angle within allowable limits:

$$\delta \leftarrow clamp(\delta, -\delta_{max}, \delta_{max}).$$

10. Predict the velocity based on acceleration and deceleration limits:

$$- v_{acc} \leftarrow v_{robot} + a \cdot \Delta t$$

$$- v_{dec} \leftarrow \sqrt{2} \cdot d \cdot s_{remaining}$$

11. Determine the final command velocity:

$$v \leftarrow clamp(\min(v_{acc}, v_{dec}), v_{min}, v_{max})$$

12. Return the control commands:

return (δ, v)

3.7.2 Pure Pursuit Path Tracking with Dynamic Lookahead and Steering-Based Speed Control (PP-DSC)

The Pure Pursuit Path Tracking with Dynamic Lookahead and Steering-Based Speed Control (PP-DSC) algorithm enhances trajectory tracking performance by integrating two key strategies. First, it dynamically adjusts the lookahead distance based on the robot's linear velocity to ensure appropriate responsiveness and stability across different speeds. A shorter lookahead distance improves path-following precision at low speeds, while a longer lookahead distance promotes smoother motion and reduces oscillations at higher speeds, thereby maintaining tracking accuracy across varying dynamic conditions. Second, it adaptively regulates the forward velocity according to the robot's steering intensity. Reducing speed during sharp turns allows

the robot to maintain better control and reduces the risk of overshooting or deviating from the planned trajectory. Conversely, maintaining higher speeds during straight-line motion improves efficiency without compromising stability. These combined strategies enable the robot to adapt its behavior dynamically to the characteristics of the path, enhancing overall robustness, motion smoothness, and trajectory-tracking precision.

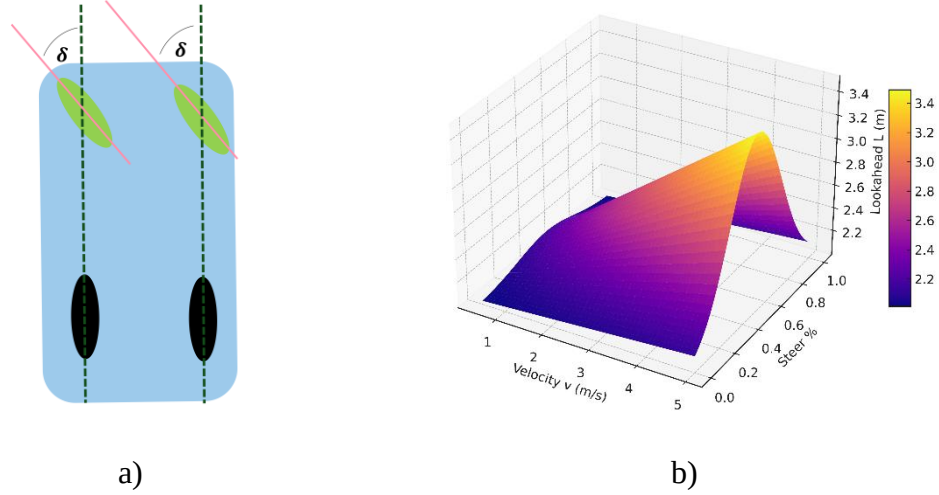


Figure 21 Steering-adaptive lookahead concept and surface plot

(a) Conceptual diagram of steering-based lookahead behavior; (b) Parametric surface of lookahead distance as a function of velocity and steering percentage.

To compute the steering angle, the algorithm follows the geometric formulation of the PP method. The steering angle δ is calculated based on the position of the lookahead point in the vehicle's local coordinate frame, as shown in Eq. (27):

$$\delta = \tan^{-1} \left(\frac{2y_{ld}L}{ld^2} \right), \quad (27)$$

In this equation, y_{ld} is the lateral displacement of the lookahead point, L is the wheelbase, and ld is the lookahead distance. To ensure that the steering remains within the robot's mechanical limits, the output from Eq. (27) is bounded using a clamping function described in Eq. (28):

$$\delta = \begin{cases} \delta_{max}, & \text{if } \delta > \delta_{max} \\ -\delta_{max}, & \text{if } \delta < -\delta_{max} \\ \delta, & \text{otherwise} \end{cases}, \quad (28)$$

The lookahead distance ld is dynamically adapted to the current robot velocity v , scaling between minimum and maximum limits. This velocity-dependent adjustment is calculated using Eq. (29):

$$ld = ld_{min} + \left(\frac{v - v_{min}}{v_{max} - v_{min}} \right) (ld_{max} - ld_{min}), \quad (29)$$

As shown in Figure 21(a), the conceptual model illustrates how steering behavior influences lookahead adjustments. Meanwhile, Figure 21(b) provides a parametric surface that highlights how the lookahead distance changes with respect to both forward velocity and steering percentage.

To achieve better stability during the turning maneuvers, the robot's velocity is modulated based on the current steering demand. This process begins by normalizing the steering angle, resulting in a steering percentage, as shown in Eq. (30):

$$steer_pct = \frac{\delta}{\delta_{max}}, \quad (30)$$

This percentage indicates how close the steering angle is to its maximum allowable value. Using this metric, the robot's forward velocity v is selected according to a piecewise function described in Eq. (31):

$$S = \frac{steer_pct - lower_steer_pct}{upper_steer_pct - lower_steer_pct}, \quad (31)$$

$$v = \begin{cases} v_{min} & \text{if } steer_pct > upper_steer_pct \\ v_{max} & \text{if } steer_pct < lower_steer_pct \\ v_{min} + (v_{max} - v_{min}) \times S & \text{otherwise} \end{cases} \quad (32)$$

By combining the steering formulation in Eq. (27), the clamping in Eq. (28), the dynamic lookahead computation in Eq. (29), and the adaptive velocity control from Eq. (30) through (32), the PP-DSC algorithm enables the robot to track reference trajectories with greater precision and smoother motion, even under varying curvature and speed conditions.

Algorithm 3: Pure Pursuit Path Tracking with Dynamic Lookahead and Steering-Based Speed Control (PP-DSC)

Input: Robot state (x, y, θ) , path P , wheelbase L , lookahead range (ld_{min}, ld_{max}) , velocity limits (v_{min}, v_{max}) , steering percentage thresholds $(upper_pct, lower_pct)$.

Output: Steering angle (δ) , linear velocity (v)

1. Acquire the current robot state:

$(x_{robot}, y_{robot}, \theta_{robot})$

2. Identify the index of the nearest waypoint on the path:

$$nearest_idx \leftarrow \underset{i}{\operatorname{argmin}} \operatorname{distance}(P[i], (x_{robot}, y_{robot}))$$

3. Search for the lookahead point:

For each point $P[i]$ from $nearest_idx$ onward:

If $\operatorname{distance}(P[i], (x_{robot}, y_{robot})) \geq ld$,

then assign $lookahead_point \leftarrow P[i]$ and break the loop.

4. Transform the $lookahead_point$ into the robot's local (base_link) frame:

$(x_{ld}, y_{ld}) \leftarrow$ Transform to robot frame

$(lookahead_point, x_{robot}, y_{robot}, \theta_{robot})$.

5. Compute the steering angle using the Pure Pursuit formula:

$$\delta \leftarrow \operatorname{atan2}(2 \cdot y_{ld} \cdot L, ld^2)$$

6. Compute the steering angle using the Pure Pursuit formula:

$$\delta \leftarrow \operatorname{clamp}(\delta, -\delta_{max}, \delta_{max})$$

7. Clamp the steering angle within allowable limits:

$$steer_pct \leftarrow \frac{|\delta|}{\delta_{max}}$$

8. Determine the linear velocity based on $steer_pct$:

If $steer_pct > upper_pct$:

$$v \leftarrow v_{min}$$

Else If $steer_pct > upper_pct$:

$$v \leftarrow v_{max}$$

Else:

$$v \leftarrow v_{min} + (v_{min} - v_{max}) \cdot \frac{upper_pct - steer_pct}{upper_pct - lower_pct}$$

9. control commands:

return (δ, v)

Table 5 Algorithm parameter configuration.

PP		PP-DSC	
Parameters	Value	Parameters	Value
L	1.10	L	1.10
ld	5.0	ld_{min}	0.5
v	5.0	ld_{max}	5.0
		v_{min}	0.5
		v_{max}	3.0
		$lower_{steer_pct}$	0.2
		$upper_{steer_pct}$	0.7

3.8 Implementation procedure

The overall autonomous navigation system developed in this study integrates positioning sensors and computational algorithms into a structured architecture, as illustrated in Figure 22. The system comprises three main components:

- **Positioning Sensors:** This block includes GNSS RTK modules, IMU modules, and encoder modules. The GNSS RTK modules provide high-accuracy satellite-based positioning data, while the IMU modules offer inertial information regarding orientation and velocity. The encoder modules track wheel rotations or motor positions to support precise motion estimation.
- **Computer and Embedded Systems:** This block consists of the computational platform that executes autonomous driving-based algorithms. These algorithms are categorized into three main types:
 - **Position Algorithm:** Processes raw sensor data to determine the current position of the robot.
 - **Localization Algorithm:** Integrates data from multiple sensors to ascertain the robot's location within a predefined coordinate framework or map.
 - **Navigation Algorithm:** Utilizes position and localization outputs to plan, control, and execute the robot's movement paths.
- **GNSS Base Station:** Depicted outside the main autonomous driving block, the GNSS base station provides differential correction signals to the GNSS

RTK modules. This correction mechanism significantly improves the positioning accuracy required for precise navigation tasks.

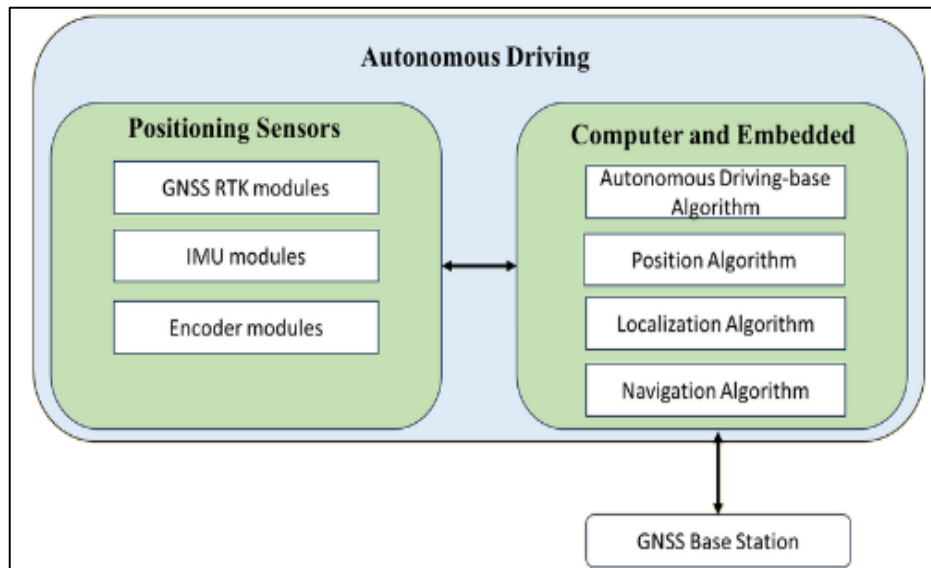


Figure 22 Diagram illustrating the components of an autonomous driving system.

The proposed navigation system was implemented following a structured workflow, as depicted in Figure 23. The procedure begins with system initialization, including the verification of sensor functionality and communication links. Once operational integrity is confirmed, localization is performed by fusing data from the GNSS RTK receiver, IMU, and wheel encoders. A state estimation module integrates these sensor inputs to generate a reliable pose estimation, which supports subsequent path planning and path-following tasks.

The system was tested in an open-sky environment under optimal GNSS conditions, as shown in Figure 24. The test field provided a flat and unobstructed area measuring approximately 64×20 meters, offering ideal conditions for evaluating the navigation system's baseline performance.

To comprehensively assess the tracking capabilities, three different trajectory types were evaluated:

- Linear Path: Representing straightforward, high-speed motion.
- Loop Path: Representing continuous, stable curving maneuvers.

- Figure-Eight Path: Representing alternating turn sequences requiring dynamic steering adjustments.

These test paths, illustrated in Figure 25, were selected to reflect a range of motion dynamics and allowed for the assessment of the system's adaptability and tracking accuracy under various navigation scenarios

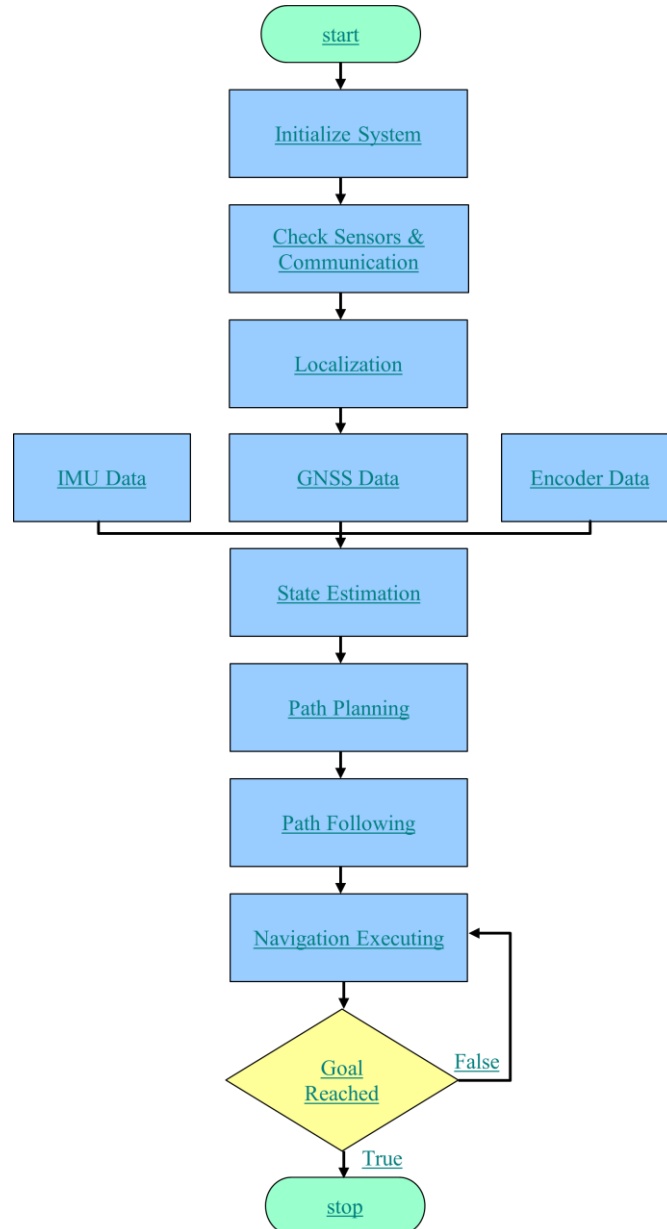


Figure 23 Flowchart of the Path Planning and Tracking Process for an Autonomous Robot



Figure 24 Field Test Location for Navigation Algorithms.

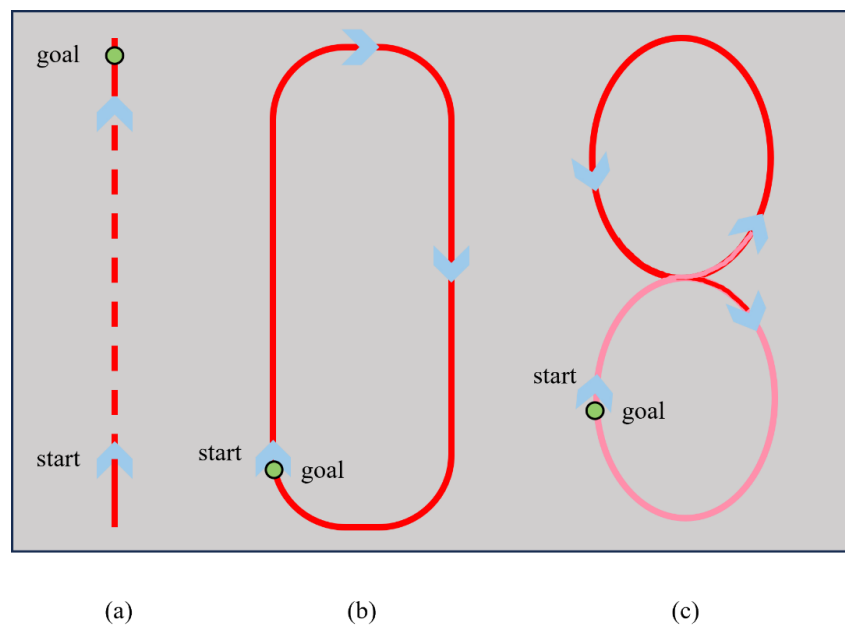


Figure 25 Test Trajectories for Robot Navigation. (a) Linear, (b) Loop, (c) Figure-Eight

CHAPTER IV

RESULTS AND DISCUSSION

4.1 Experiment Setting

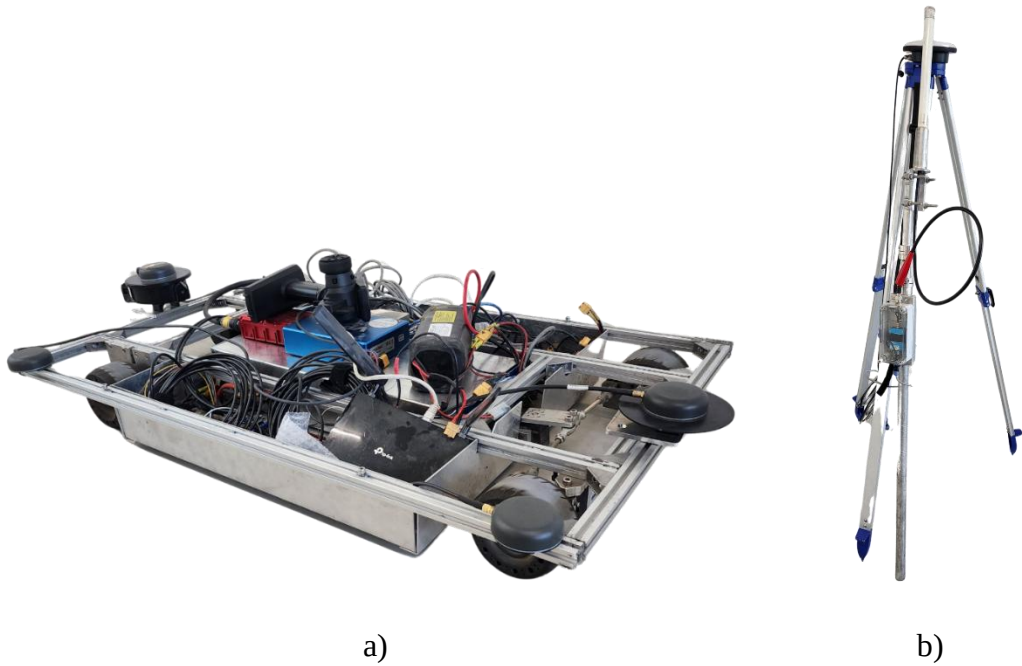


Figure 26 Experimental platform and GNSS-RTK base station setup
(a) Mobile robot platform equipped with GNSS, IMU, and onboard computer
(b) GNSS-RTK base station on tripod for high-precision localization

To evaluate the performance of the proposed GNSS-RTK-based navigation system, a series of field experiments were conducted to assess the robot's path-tracking accuracy and positioning capability under real-world conditions. The test environment, illustrated in Figure 26, was an open field with minimal GNSS signal obstruction, providing stable conditions for localization. The experiments were structured around three types of trajectories: a linear path, a loop path, and a figure-eight path, with the robot operating at speeds ranging from 1 to 5 m/s. Operational routes were defined using UTM coordinates to enable precise control within a Cartesian reference frame. The experimental workflow followed a standardized navigation sequence, which

included system initialization, sensor fusion for localization, path planning, and autonomous execution.

Two tracking algorithms were evaluated during the experiments. The standard Pure Pursuit (PP) algorithm was employed as the baseline method for comparison, while the enhanced Pure Pursuit with Dynamic Lookahead and Steering-Based Speed Control (PP-DSC) algorithm was tested to assess its improvements. The PP-DSC algorithm incorporated a dynamic lookahead mechanism and steering-aware velocity modulation to enhance tracking accuracy and stability, particularly at higher speeds and during curved maneuvers. The evaluation primarily focused on trajectory adherence and positioning precision, using GNSS-RTK data as the main localization source. Experimental conditions are summarized in Table 6, and all tests were conducted in real-time outdoor environments to validate the practical applicability and robustness of the proposed system.

Table 6 Summary of experimental conditions.

Parameter	Value
Evaluation Metrics	Path tracking accuracy, positioning capability
Coordinate System	UTM (Universal Transverse Mercator)
Control Format	Cartesian (x, y)
Test Scenarios	line path, loops path, figure-eight path
Speed Range	1 - 5 m/s
Positioning Method	GNSS-RTK
Environment	Real-world conditions (actual routes)

4.2 Performance comparison of path tracking algorithms

To assess the effectiveness of the proposed PP-DSC algorithm, a comparative evaluation was conducted against the conventional Pure Pursuit (PP) algorithm using three reference trajectories: straight-line, circular, and figure-eight paths, selected to

examine the algorithms' capability in handling directional variability and curvature transitions.

For the straight-line trajectory (Figure 27(a)), the PP algorithm with a fixed lookahead distance of 4 meters resulted in an RMSE of 0.22 meters and a maximum lateral deviation of 0.41 meters, whereas PP-DSC reduced the RMSE to 0.05 meters and the maximum deviation to 0.09 meters. Oscillations during the initial motion phase under the PP algorithm, attributed to short lookahead settings, align with prior findings [58].

Regarding the circular trajectory (Figure 27(b)), PP produced an RMSE of 0.29 meters and a maximum deviation of 0.44 meters. In contrast, PP-DSC, by dynamically adjusting lookahead distance and velocity before curve entry, achieved an RMSE of 0.07 meters and a maximum deviation of 0.11 meters, thus improving path conformity.

For the figure-eight trajectory (Figure 27(c)), PP exhibited frequent overshoot and undershoot with an RMSE of 0.24 meters and a maximum deviation of 0.34 meters, whereas PP-DSC maintained higher stability, achieving an RMSE of 0.08 meters and capping deviation at 0.13 meters. These results corroborate previous studies [59], confirming the advantages of dynamic lookahead adjustment under rapidly changing curvature profiles.

Quantitative performance across all trajectories is presented in Table 7 and visualized in Figure 28. The results clearly indicate that PP-DSC outperformed the standard PP algorithm across all error metrics, including mean deviation, RMSE, mean absolute error (MAE), and maximum deviation. When considered alongside prior studies [58-59], the findings confirm that dynamically adapting the lookahead distance and velocity in response to real-time steering conditions plays a critical role in reducing oscillations, enhancing stability, and achieving high-precision path tracking in diverse navigation scenarios.

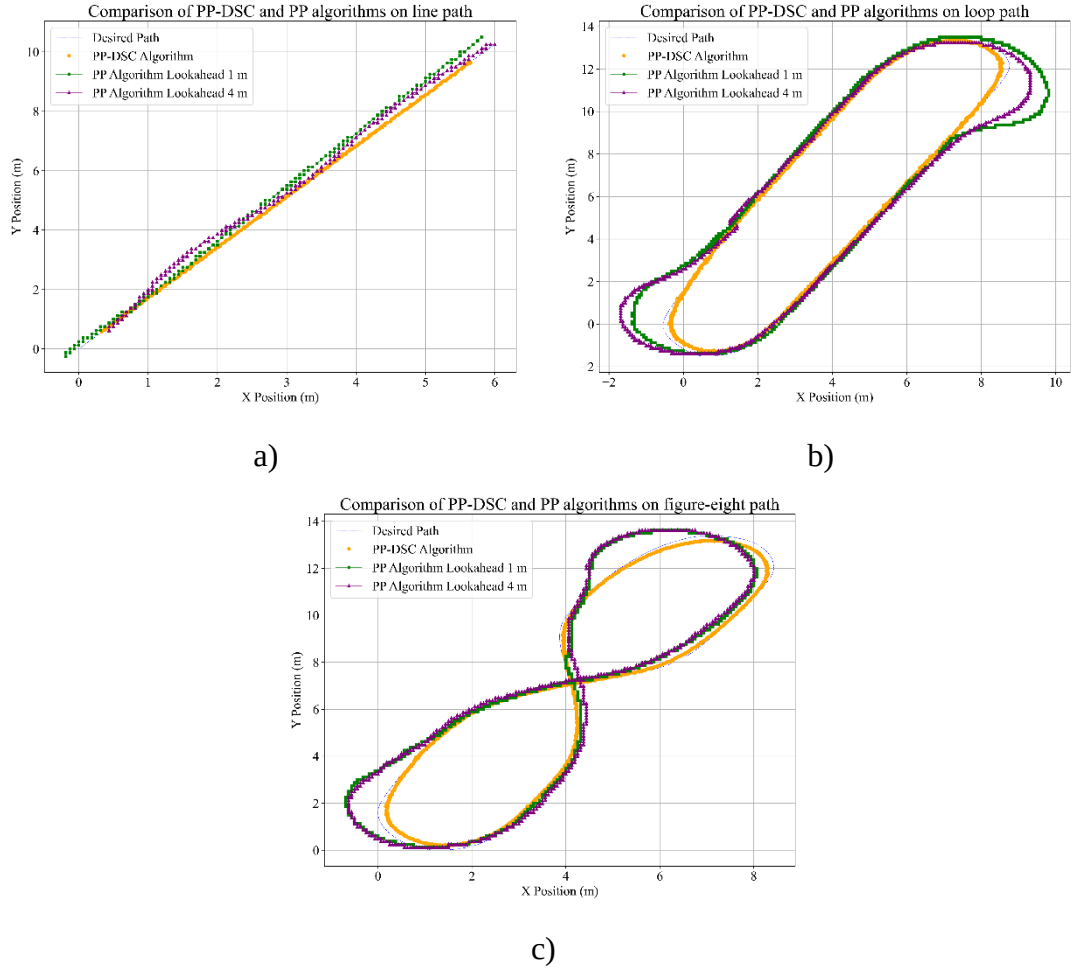


Figure 27 PP-DSC vs PP performance on test trajectories: (a) Line;(b) Loop;(c) Figure-eight).

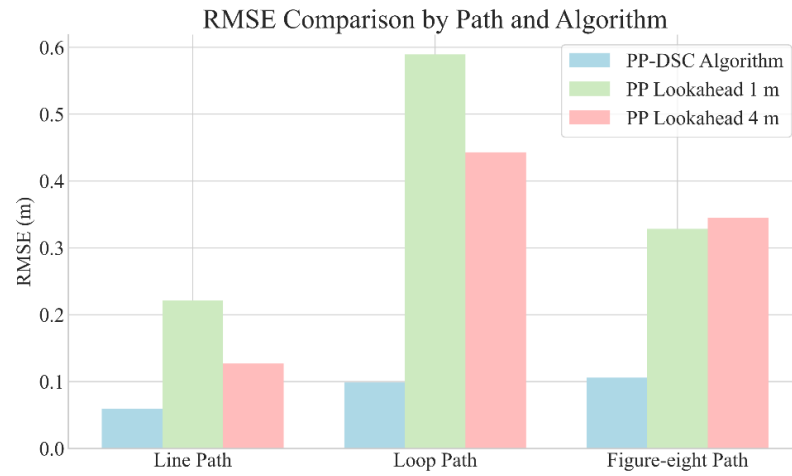


Figure 28 RMSE performance of PP-DSC and PP across test trajectories.

Table 7 Performance comparison of PP-DSC and PP algorithms on various path types

Tracking path	Algorithm	Look ahead (m)	Mean (m)	RMSE (m)	MAE (m)	Max error (m)	STD (m)
Line	PP	1	0.11	0.12	0.11	0.30	0.06
	PP	4	0.19	0.22	0.19	0.31	0.09
	PP-DSC	Dyna mic	0.05	0.05	0.05	0.09	0.02
Loop	PP	1	0.28	0.44	0.28	1.31	0.33
	PP	4	0.40	0.58	0.40	1.59	0.43
	PP-DSC	Dyna mic	0.07	0.09	0.07	0.30	0.06
Figure-eight	PP	1	0.27	0.34	0.27	0.93	0.21
	PP	4	0.25	0.32	0.25	0.89	0.20
	PP-DSC	Dyna mic	0.08	0.10	0.08	0.26	0.06

4.3 Speed Adaptation Behavior

The ability to regulate a robot's velocity based on path geometry is critical for enhancing tracking accuracy and motion stability in autonomous navigation. This capability becomes particularly important when the robot encounters directional changes or varying curvature along the trajectory. The conventional Pure Pursuit (PP) algorithm employs a fixed lookahead distance and does not adapt the velocity in real time. In contrast, the proposed Pure Pursuit with Dynamic Lookahead and Steering-Based Speed Control (PP-DSC) algorithm dynamically modulates the robot's velocity according to the steering percentage, which is calculated as the ratio between the current steering angle and the maximum allowable steering angle, as defined in Eq. (30). The velocity is subsequently adjusted within predefined upper and lower bounds using Eqs. (31) and (32), enabling the robot to decelerate during sharp turns and accelerate along straight segments.

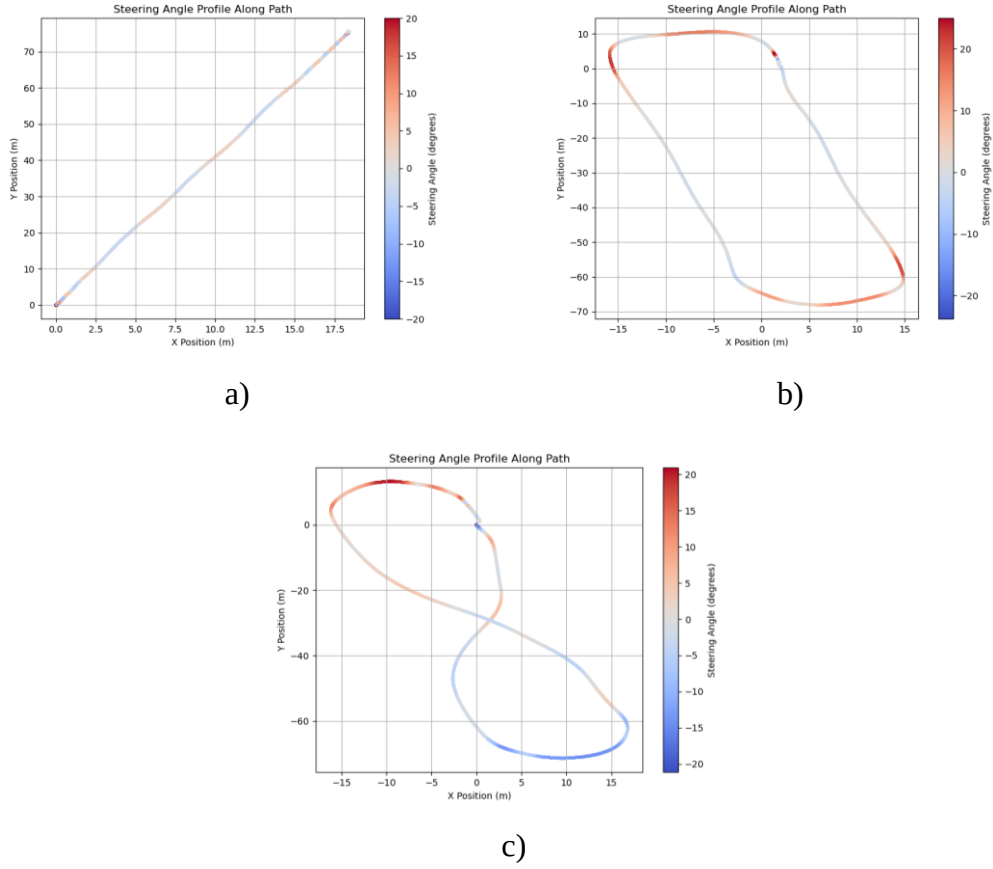
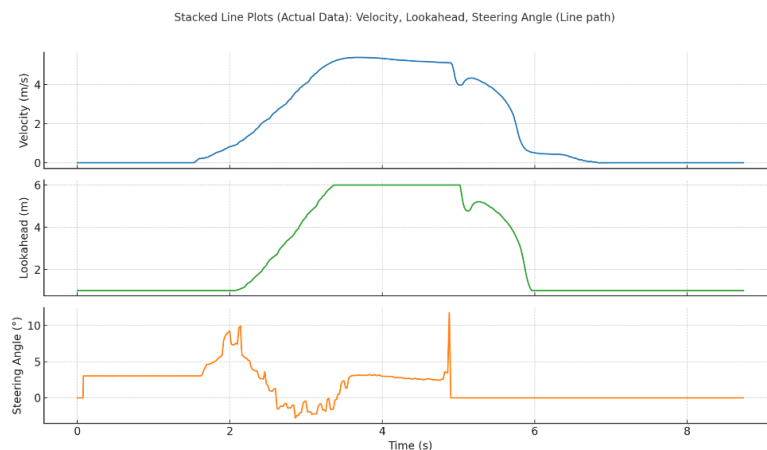


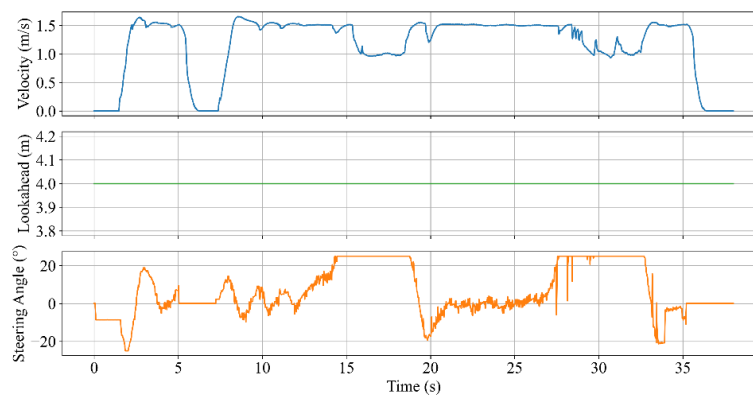
Figure 29 Steering angle distributions along different trajectory types: (a) Straight-Line Path, (b) Loop Path and (c) Figure-Eight Path

The steering angle profiles corresponding to different trajectories are depicted in Figure 29. On the straight-line trajectory (Figure 29(a)), the steering angles remained close to zero, indicating minimal directional deviation and allowing the robot to sustain a higher and more stable velocity. On the loop trajectory (Figure 29(b)), greater variations in steering angles were observed, necessitating dynamic speed adjustments to maintain path-tracking stability. Specifically, the robot decelerated while navigating curved segments and reaccelerated on straighter portions of the path.



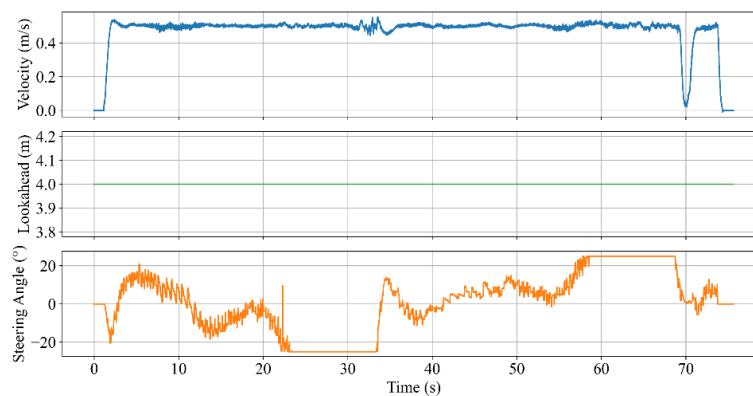
a)

Stacked line plots : velocity, lookahead and steering angle of loop path



b)

Stacked line plots : velocity, lookahead and steering angle of figure-eight path



c)

Figure 30 Velocity, lookahead distance, and steering angle on different path types: (a) line; (b) loop; (c) figure-eight.

For the figure-eight trajectory (Figure 29(c)), frequent and sharp transitions between positive and negative steering angles were evident, reflecting continuous directional changes that required repeated speed modulation to ensure smooth and accurate trajectory tracking. These findings demonstrate that integrating steering feedback with velocity control, as realized in the PP-DSC algorithm, is essential for achieving adaptive, smooth, and robust navigation performance across a variety of dynamic path conditions.

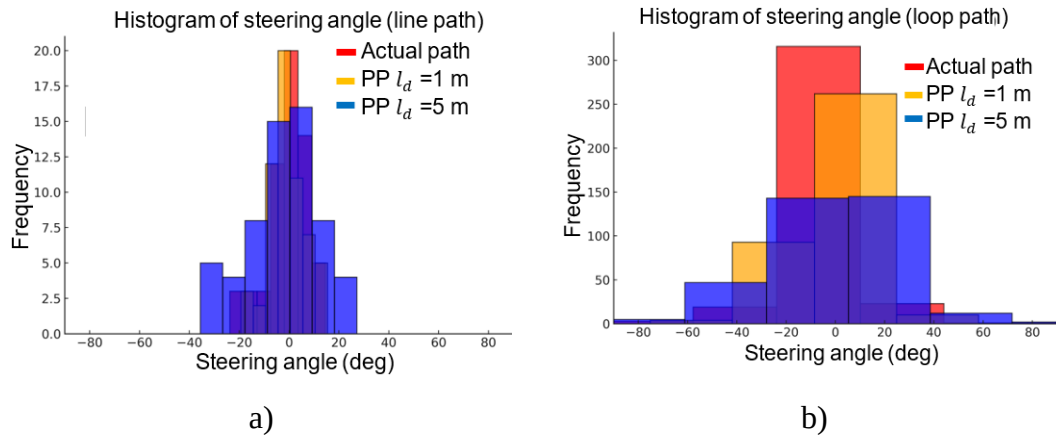


Figure 31 Steering angle histogram: line path (a), loop path (b).

In the straight-line test, steering angles remained close to zero throughout the trajectory, which is consistent with expectations for a path with no curvature (Figure 31(a)). The actual robot data and both simulated cases using 1-meter and 5-meter lookahead distances showed similar results. Most values clustered tightly between -5 and $+5$ degrees, with minimal spread. Since the path did not require significant turning, all methods produced nearly identical steering behavior.

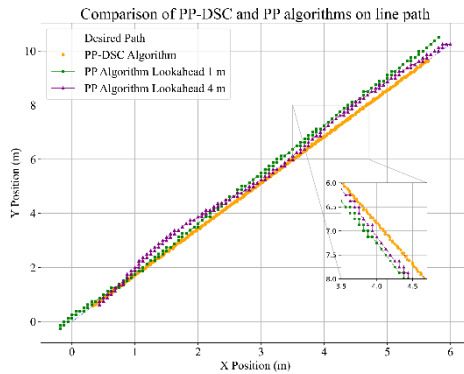
The loop path, which includes continuous curved segments, resulted in a wider range of steering angles (Figure 31(b)). The actual steering values were concentrated mostly between -20 and $+20$ degrees. When comparing the two lookahead settings, the 1-meter configuration showed a narrower distribution, while the 5-meter version introduced more variance and occasional steering commands that extended beyond ± 30 degrees. This suggests that the longer lookahead led to less responsive adjustments, especially in curve entry and exit zones.

Table 8 Comparison of speed performance and smoothness for PP-DSC and PP algorithms

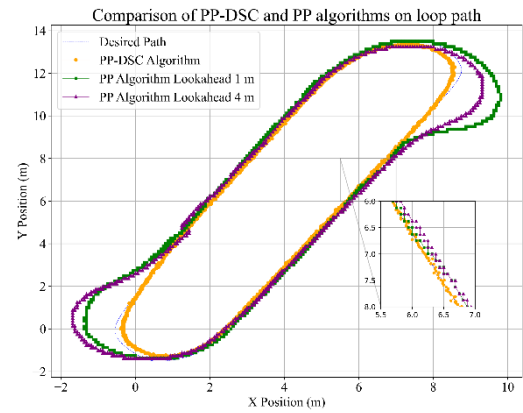
Path	Algorithm	Lookahead	Avg Speed (m/s)	Max Speed (m/s)	Min Speed (m/s)	Std Dev (m/s)	Speed Drop in Turns	Smooth Score
Line	PP	1.0	3.11	4.87	0.5	2.32	0.3	1.0
	PP	4.0	2.41	4.82	0.5	2.41	0.1	1.0
	PP-DSC	Dynamic	4.39	4.62	4.62	0.2	0.8	4.2
Loop	PP	1.0	2.51	3.0	0.5	0.98	0.3	1.08
	PP	4.0	2.6	3.0	0.5	1.02	0.1	1.0
	PP-DSC	Dynamic	2.1	3.0	0.5	1.16	2.5	1.0
Figure-eight	PP	1.0	2.13	2.78	0.5	1.15	0.3	1.0
	PP	4.0	2.64	2.75	0.5	0.54	0.1	2.84
	PP-DSC	Dynamic	0.99	3.0	0.5	1.03	2.5	1.0

4.4 Localization Accuracy

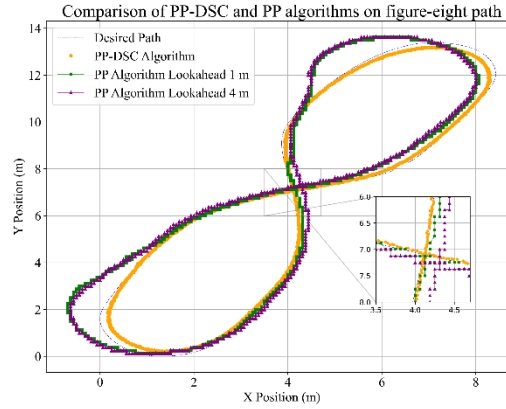
Localization accuracy is a critical metric for evaluating the effectiveness of path-tracking algorithms, particularly in systems that integrate GNSS-RTK with real-time motion control. In this study, accuracy was assessed by comparing the robot's actual position, converted into UTM coordinates using Eq. (5), against the reference trajectory. Positional deviations at each time step were used to compute standard spatial error metrics, including mean error, root mean square error (RMSE), and maximum error.



a)



b)



c)

Figure 32 Path tracking comparison with zoomed-in segments: (a) line; (b) loop; (c) figure-eight.

The experimental results indicated that the standard Pure Pursuit (PP) algorithm exhibited lower localization accuracy, particularly during curved segments and rapid directional changes. Significant deviations were observed along the loop path between 18 and 24 meters and at the intersection region of the figure-eight path between 15 and 25 meters, as shown in Figure 31. For the straight path (Figure 32(a)), the PP algorithm with a 4-meter lookahead distance produced an RMSE of 0.22 meters and a maximum error of 0.41 meters. In contrast, the PP-DSC algorithm improved accuracy, achieving an RMSE of 0.05 meters and a maximum error of 0.09 meters.

On the loop trajectory (Figure 32(b)), the PP algorithm exhibited overshooting behavior in the left-hand curve section between 18 and 24 meters, resulting in an RMSE of 0.29 meters and a maximum error of 0.44 meters. The PP-DSC algorithm significantly reduced localization errors in this segment, lowering the RMSE to 0.07 meters and the maximum error to 0.11 meters. For the figure-eight trajectory (Figure 32(c)), characterized by alternating left and right curves, particularly between 15 and 25 meters, the PP algorithm demonstrated inconsistent tracking performance with an RMSE of 0.24 meters and a maximum error of 0.34 meters. In comparison, the PP-DSC algorithm maintained more consistent performance, achieving an RMSE of 0.08 meters and a maximum error of no more than 0.13 meters.

Table 9 Deviation metrics for PP and PP-DSC algorithms across different path types

Path	Algorithm	Lookahead(m)	Avg deviation	Max deviation	Std deviation
Line	PP	1	0.93	2.89	1.15
	PP	4	0.19	0.75	0.22
	PP-DSC	Dynamic	0.05	0.09	0.02
Loop	PP	1	0.41	1.5	0.41
	PP	4	0.28	1.31	0.28
	PP-DSC	Dynamic	0.07	0.30	0.06
Figure-eight	PP	1	0.25	0.89	0.20
	PP	4	0.27	0.93	0.21
	PP-DSC	Dynamic	0.08	0.26	0.06

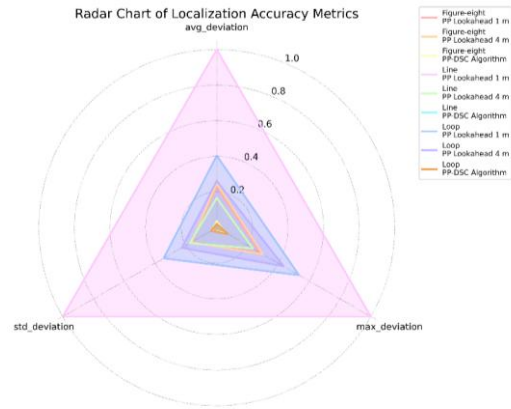
**Figure 33** Radar chart of localization deviation metrics for PP and PP-DSC algorithms.

Table 10 Summary of observations from PP-DSC field testing

Path	Speed (m/s)	Observations	Error (%)
Line	5	Stable tracking, minimal steering adjustments required	0.4
Loop	5	Reduced deviations at turns due to dynamic steering	0.9
Figure-eight	5	Improved handling in complex curves, reduced sharp-turn errors	1.3

Quantitative results presented in Table 9 and illustrated in Figure 33 clearly demonstrate the superior localization accuracy of the PP-DSC algorithm across all tested paths. Observations from the field trials, summarized in Table 10, further confirm that the PP-DSC algorithm maintained stable positional accuracy during straight-line movement and effectively minimized deviations during curved segments. When analyzing the results by path segment type, as shown in Table 11, it becomes evident that the PP algorithm exhibited larger deviations during curved paths, whereas the PP-DSC algorithm consistently maintained low and stable error in both straight and curved segments. Spatial plots provided in Figure 34 visualize the actual robot trajectories compared to the reference paths, reaffirming that the PP-DSC algorithm achieved higher localization precision and consistency under all experimental conditions.

Table 11 Quantitative error comparison between PP and PP-DSC

Path	Algorithm	Lookahead	Segment type	Mean deviation (m)	Max deviation (m)	Std deviation (m)
Line	PP	1	Straight	1.07	2.89	1.22
		1	Curve	0.44	2.89	0.67
		4	Straight	0.20	0.75	0.24
		4	Curve	0.15	0.72	0.10
	PP-DSC	Dynamic	Straight	0.05	0.09	0.02
		Dynamic	Curve	0.05	0.09	0.03
Loop	PP	1	Straight	0.42	1.59	0.42
		1	Curve	0.33	1.59	0.40
		4	Straight	0.28	1.32	0.27
		4	Curve	0.28	1.32	0.32
	PP-DSC	Dynamic	Straight	0.05	0.22	0.05
		Dynamic	Curve	0.07	0.30	0.07
Figure-eight	PP	1	Straight	0.25	0.89	0.20
		1	Curve	0.25	0.89	0.24
		4	Straight	0.27	0.94	0.21
		4	Curve	0.28	0.94	0.24
	PP-DSC	Dynamic	Straight	0.07	0.24	0.06
		Dynamic	Curve	0.08	0.26	0.07

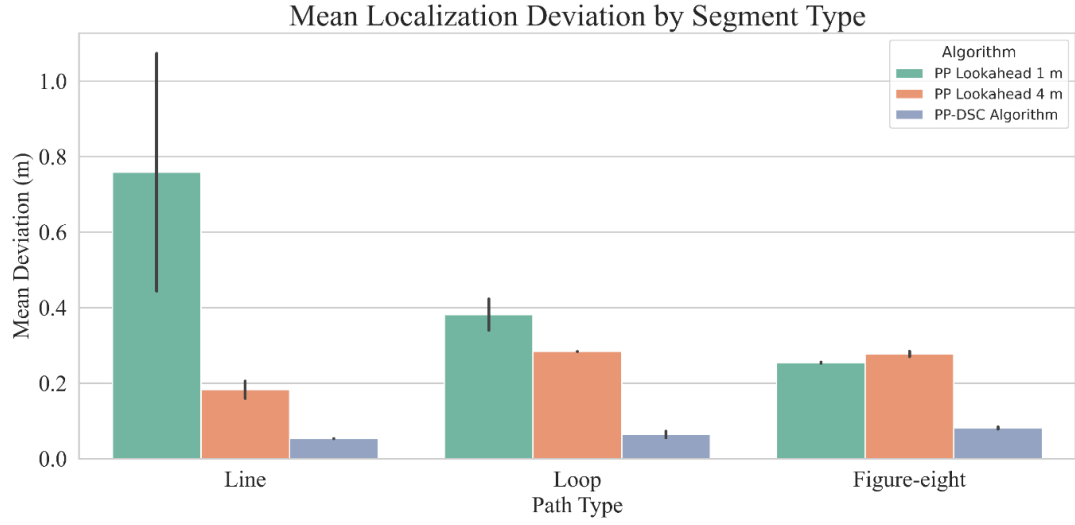


Figure 34 Localization deviation comparison highlighting the consistency of PP-DSC across path types.

In autonomous navigation, the interplay between linear velocity, angular velocity, and turning radius plays a critical role, as illustrated in Figure 35. The relationship among these parameters fundamentally determines the maneuverability and stability of a mobile robot during path tracking. The turning radius r can be expressed as the ratio of the linear velocity v to the angular velocity ω , formulated by the equation $r = \frac{v}{\omega}$.

As shown in the figure, for a given angular velocity, increasing the linear velocity results in a larger turning radius, meaning the robot follows a gentler curve. Conversely, for a fixed linear velocity, increasing the angular velocity leads to a smaller turning radius, enabling the robot to execute sharper turns.

This interaction highlights the need for coordinated control of both linear and angular velocities to maintain desired path curvature, especially when adapting to varying path geometries. In straight-line motion, where ω approaches zero, the turning radius tends toward infinity, allowing the robot to maintain high speeds. In contrast, when navigating tight curves requiring high angular velocities, the linear velocity must often be reduced to achieve a smaller, controllable turning radius without sacrificing path-following accuracy or causing instability.

Understanding and managing this dynamic relationship is essential for designing navigation algorithms such as Pure Pursuit with Dynamic Lookahead and Steering-Based Speed Control (PP-DSC), which leverage adjustments in velocity profiles based on steering behavior to optimize tracking performance under diverse operational conditions.

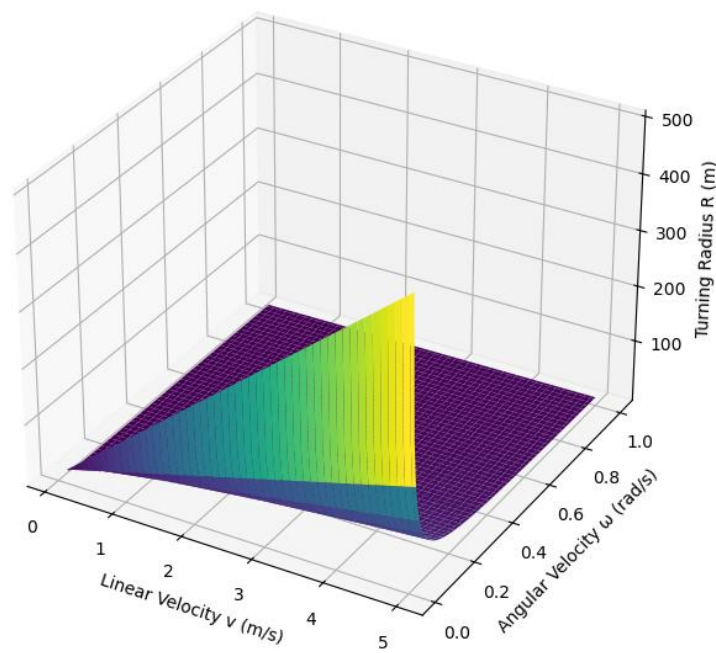


Figure 35 Surface plot of the relationship between linear velocity, angular velocity, and turning radius in autonomous robot navigation.

CHAPTER V

CONCLUSION

This study presented the development of an advanced navigation framework for steering-type autonomous mobile robots, introducing an enhanced version of the Pure Pursuit (PP) algorithm known as Pure Pursuit with Dynamic Steering Control (PP-DSC). The system was implemented on a ROS 2 platform and integrated multiple sensors including GNSS-RTK, IMU, and wheel encoders to achieve high-precision localization and robust trajectory tracking under various path conditions.

The primary conclusions drawn from the study's findings reveal several key characteristics associated with autonomous navigation performance:

- The PP-DSC algorithm dynamically adjusts the lookahead distance and linear velocity based on the steering demand, resulting in improved path-tracking accuracy compared to the conventional PP algorithm.
- The enhanced algorithm reduced the mean lateral deviation to 0.05 meters on straight paths and maintained deviations below 0.08 meters on curved paths, outperforming the maximum deviation of 0.44 meters observed with standard PP.
- The dynamic adjustment strategy effectively stabilized the robot's motion during complex trajectories such as loops and figure-eight patterns.
- The system achieved reliable performance at speeds up to 5.0 meters per second, demonstrating real-time localization robustness even under varied field conditions.

These results emphasize the importance of adaptive control strategies in improving both motion stability and trajectory tracking precision. The success of the PP-DSC algorithm confirms that real-time adjustments of control parameters in response to the robot's dynamics are essential for dependable autonomous navigation, particularly in outdoor agricultural applications.

This research proposes a practical and effective approach for improving the performance of autonomous navigation systems, especially in contexts where high positioning accuracy and real-time responsiveness are critical. While this study was

conducted using a mobile robot prototype, the proposed framework is highly adaptable and can be applied to various types of agricultural machinery such as autonomous tractors, tillers, and field robots. This can improve operational efficiency, reduce labor requirements, and support the advancement of precision agriculture.

Moreover, the developed PP-DSC algorithm holds strong potential for application in agricultural robotics. Its ability to dynamically adjust the lookahead distance and linear velocity allows autonomous systems to maintain precise trajectories even in narrow crop rows or structured field environments. This makes it highly suitable for use in tasks such as autonomous spraying, seeding, harvesting, or row-following operations in orchards and vineyards, where accuracy and consistency are essential to avoid crop damage and improve operational efficiency. The algorithm's modular design also allows it to be integrated into a wide range of mobile platforms used in precision agriculture.

Future development of the system may focus on expanding the algorithm's capabilities to support obstacle avoidance, refining parameter tuning for high-speed operations, and incorporating machine learning techniques to improve adaptability in dynamic and unpredictable environments. These enhancements would contribute to the continued advancement of autonomous systems in both agricultural and industrial applications. Although the current framework has demonstrated reliable performance in flat, planar environments, it has not yet been tested in terrains with elevation changes or irregular surfaces such as slopes, hills, or rugged agricultural fields. In such conditions, additional factors including gravitational effects, slip dynamics, and tilt-induced localization errors may significantly influence the control performance and stability of the robot. Nevertheless, the system possesses strong potential for further development toward slope-capable navigation. With the integration of terrain inclination estimation, slope-aware path planning, and tilt-compensated control mechanisms, the framework could be extended to achieve precise and stable navigation across a wide range of real-world topographies.

In conclusion, the research successfully achieves its original objectives by delivering an intelligent control system that enables accurate positioning and responsive maneuvering, along with a robust navigation framework that supports real-time autonomous operation in dynamic and wide-open environments.

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